

15

Nucleophilic substitution at saturated carbon

Connections

➡ Building on

- Attack of nucleophiles on carbonyl groups [ch6 & ch9](#)
- Substitution at carbonyl groups [ch10](#)
- Substitution of the oxygen atom of carbonyl groups [ch11](#)
- Reaction mechanisms [ch12](#)
- ^1H NMR [ch13](#)
- Stereochemistry [ch14](#)

Arriving at

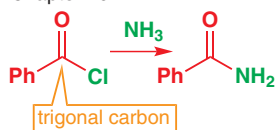
- Nucleophilic attack on saturated carbon atoms, leading to substitution reactions
- How substitution at a saturated carbon atom differs from substitution at $\text{C}=\text{O}$
- Two mechanisms of nucleophilic substitution
- Intermediates and transition states in substitution reactions
- How substitution reactions affect stereochemistry
- What sort of nucleophiles can substitute, and what sort of leaving groups can be substituted
- The sorts of molecules that can be made by substitution, and what they can be made from

➡ Looking forward to

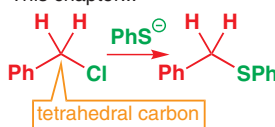
- Elimination reactions [ch17](#)
- Substitution reactions with aromatic compounds as nucleophiles [ch21](#)
- Substitution reactions with enolates as nucleophiles [ch25](#)
- Retrosynthetic analysis [ch28](#)
- Participation, rearrangement, and fragmentation reactions [ch36](#)

Mechanisms for nucleophilic substitution

Chapter 10...



This chapter...



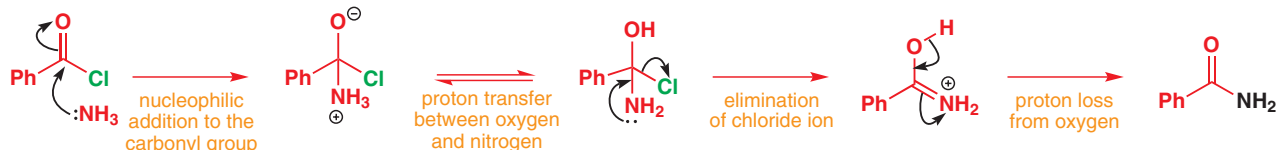
Substitution is the replacement of one group by another. You met such reactions in Chapter 10, and an example is shown in the margin. This reaction is a substitution because the Cl group is replaced by the NH_2 group. You learnt to call the molecule of ammonia (NH_3) the **nucleophile** and the chloride you called the **leaving group**. In Chapter 10, the substitution reactions always took place at the *trigonal* (sp^2) carbon atom of a carbonyl group.

In this chapter we shall be looking at reactions such as the second reaction in the margin. These are substitution reactions, because the Cl group is replaced by the PhS group. But the CH_2 group at which the reaction takes place is a *tetrahedral* (sp^3), or *saturated*, carbon atom, rather than a $\text{C}=\text{O}$ group. This reaction and the one above may look superficially the same but they are quite different in mechanism. The requirements of good reagents are also different in substitutions at carbonyl groups and at saturated carbon—that's why we changed the nucleophile from NH_3 to PhS^- : ammonia would not give a good yield of PhCH_2NH_2 in the second reaction.

Let's have a look at why the mechanisms of the two substitutions must be different. Here's a summary of the mechanism of the first reaction.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

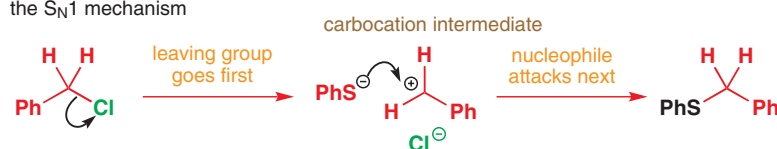
mechanism of nucleophilic substitution at the carbonyl group



In the first step the nucleophile attacks the $\text{C}=\text{O}$ π bond. It's immediately obvious that the first step is no longer possible at a saturated carbon atom. The electrons cannot be added to a π bond as the CH_2 group is fully saturated. In fact there is no way for the nucleophile to add before the leaving group departs (as it did in the reaction above) because this would give an impossible five-valent carbon atom.

Instead, two new and different mechanisms become possible. Either the leaving group goes first and the nucleophile comes in later, or the two events happen at the same time. The first of these possibilities you will learn to call the $\text{S}_{\text{N}}1$ mechanism. The second mechanism, which shows how the neutral carbon atom can accept electrons provided it loses some at the same time, you will learn to call the $\text{S}_{\text{N}}2$ mechanism. You will see later that both mechanisms are possible with this molecule, benzyl chloride.

the $\text{S}_{\text{N}}1$ mechanism



the $\text{S}_{\text{N}}2$ mechanism



Interactive mechanism for amide formation

Interactive mechanisms for $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$

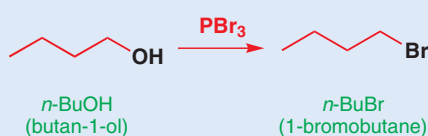
Why is it important to know about the two mechanisms for substitution?

If we know which mechanism a compound reacts by, we know what sort of conditions to use to get good yields in substitutions. For example, if you look at a commonly used nucleophilic substitution, the replacement of OH by Br , you'll find that two quite different reaction conditions are used depending on the structure of the alcohol. *Tertiary* alcohols react rapidly with HBr to give tertiary alkyl bromides. *Primary* alcohols, on the other hand, react only very slowly with HBr and are usually converted to primary alkyl bromides with PBr_3 . The reason is that the first example is an $\text{S}_{\text{N}}1$ reaction while the second is an $\text{S}_{\text{N}}2$ reaction: by the end of this chapter you will have a clear picture of how to predict which mechanism will apply and how to choose appropriate reaction conditions.

substitution of a tertiary alcohol



substitution of a primary alcohol



For $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ the reaction is **second order** (its rate depends on both $[\text{R-Br}]$ and $[\text{OH}^-]$)

For $(\text{CH}_3)_3\text{CBr}$ the reaction is **first order** (its rate depends only on $[\text{R-Br}]$ and not on $[\text{OH}^-]$)

Kinetic evidence for the $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ mechanisms

Before we go any further we are going to look in a bit more detail at these two mechanisms because they allow us to explain and predict many aspects of substitution reactions. The evidence that convinced chemists that there are two different mechanisms for substitution at saturated carbon is kinetic: it relates to the rate of reactions such as the displacement of bromide by hydroxide, as shown in the margin.

It was discovered, chiefly by Hughes and Ingold in the 1930s, that some nucleophilic substitutions are first order (that is, the rate depends only on the concentration of the alkyl halide

Edward David Hughes (1906–63) and Sir Christopher Ingold (1893–1970) worked at University College, London in the 1930s. They first thought of many of the mechanistic ideas that chemists now take for granted.

■ There is more about the relationship between reaction rates and mechanisms in Chapter 12. Quantities in square brackets represent concentrations and the proportionality constant k is called the rate constant.

■ Please note how this symbol is written. The S and the N are both capitals and the N is a subscript.

and does not depend on the concentration of the nucleophile), while others are second order (the rate depends on the concentrations of both the alkyl halide and the nucleophile). How can we explain this result? In what we called the ' S_N2 mechanism' on p. 329 there is just one step. Here's the one-step S_N2 mechanism for substitution of *n*-butyl bromide by hydroxide:



With only one step, that step must be the **rate-determining step**. The rate of the overall reaction depends only on the rate of this step, and kinetic theory tells us that the rate of a reaction is proportional to the concentrations of the reacting species:

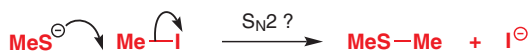
$$\text{rate of reaction} = k[\textit{n}\text{-BuBr}][\text{HO}^-]$$

If this mechanism is right, then the rate of the reaction will be simply and linearly proportional to both $[\textit{n}\text{-BuBr}]$ and $[\text{HO}^-]$. And it is. Ingold measured the rates of reactions like these and found that they were proportional to the concentration of each reactant—in other words they were second order. He called this mechanism Substitution, Nucleophilic, 2nd order; S_N2 for short. The rate equation is usually given like this, with k_2 representing the second-order rate constant.

$$\text{rate} = k_2[\textit{n}\text{-BuBr}][\text{HO}^-]$$

Significance of the S_N2 rate equation

This equation is useful for two reasons. Firstly, it gives us a test for the S_N2 mechanism. Let's illustrate this with another example: the reaction between NaSMe (an ionic solid—the nucleophile will be the anion MeS^-) and MeI to give Me_2S , dimethyl sulfide.



To study the rate equation, first, we keep the concentration of NaSMe constant and in a series of experiments vary that of MeI and see what happens to the rate. Then in another set of experiments we keep the concentration of MeI constant and vary that of MeSNa and see what happened to the rate. If the reaction is indeed S_N2 we should get a linear relationship in both cases: the graphs in the margin show a typical set of results.

The first graph tells us that the rate is proportional to $[\text{MeI}]$, that is, $\text{rate} = k_a[\text{MeI}]$ and the second graph that it is proportional to $[\text{MeSNa}]$, that is, $\text{rate} = k_b[\text{MeSNa}]$. But why are the slopes different? If you look at the rate equation for the reaction, you will see that we have incorporated a constant concentration of one of the reagents into what appears to be the rate constant for the reaction. The true rate equation is

$$\text{rate} = k_2[\text{MeSNa}][\text{MeI}]$$

If $[\text{MeSNa}]$ is constant, the equation becomes

$$\text{rate} = k_a[\text{MeI}], \text{ where } k_a = k_2[\text{MeSNa}]$$

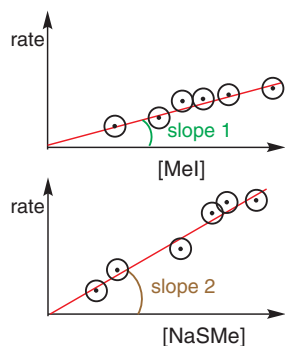
If $[\text{MeI}]$ is constant, the equation becomes

$$\text{rate} = k_b[\text{MeSNa}], \text{ where } k_b = k_2[\text{MeI}]$$

If you examine the graphs you will see that the slopes are different because

$$\text{slope 1} = k_a = k_2[\text{MeSNa}], \text{ but slope 2} = k_b = k_2[\text{MeI}]$$

We can easily measure the true rate constant k_2 from these slopes because we know the constant values for $[\text{MeSNa}]$ in the first experiment and for $[\text{MeI}]$ in the second. The value of k_2



from both experiments should be the same. The mechanism for this reaction is indeed S_N2 : the nucleophile MeS^- attacks as the leaving group I^- leaves.

The second reason that the S_N2 rate equation is useful is that it confirms that the performance of an S_N2 reaction depends **both on the nucleophile and on the carbon electrophile**. We can therefore make a reaction go better (speed it up or improve its yield) by changing either. For example, if we want to displace I^- from MeI using an oxygen nucleophile we might consider using any of those in the table below.

Oxygen nucleophiles in the S_N2 reaction

Oxygen nucleophile	$\text{p}K_a$ of conjugate acid	Rate in S_N2 reaction
HO^-	15.7 (H_2O)	fast
RCO_2^-	about 5 (RCO_2H)	moderate
H_2O	-1.7 (H_3O^+)	slow
RSO_2O^-	0 (RSO_2OH)	slow

➡ See Chapter 8 for discussion of $\text{p}K_a$ values.

The same reasons that made hydroxide ion basic (chiefly that it is unstable as an anion and therefore reactive) make it a good nucleophile. Basicity can be viewed as nucleophilicity towards a proton, and nucleophilicity towards carbon must be related. So if we want a fast reaction, we should use NaOH rather than, say, Na_2SO_4 to provide the nucleophile. Even at the same concentration, the rate constant k_2 with HO^- as the nucleophile is much greater than the k_2 with SO_4^{2-} as the nucleophile.

But that is not our only option. The reactivity and hence the structure of the carbon electrophile matter too. If we want reaction at a methyl group we can't change the carbon skeleton, but we can change the leaving group. The table below shows what happens if we use the various methyl halides in reaction with NaOH . The best choice for a fast reaction (greatest value of k_2) will be to use MeI and NaOH to give methanol.

You saw in Chapter 10 that nucleophilicity towards the carbonyl group is closely related to basicity. The same is not quite so true for nucleophilic attack on the saturated carbon atom, as we shall see, but there is a relationship nonetheless.

■ We shall discuss nucleophilicity and leaving group ability in more detail later.



Halide leaving groups in the S_N2 reaction

Halide X in MeX	$\text{p}K_a$ of conjugate acid HX	Rate of reaction with NaOH
F	+3	very slow indeed
Cl	-7	moderate
Br	-9	fast
I	-10	very fast

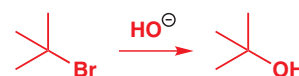
● **The rate of an S_N2 reaction depends on:**

- the nucleophile
- the carbon skeleton
- the leaving group

along with the usual factors of temperature and solvent.

Significance of the S_N1 rate equation

If we replace the substitution of *n*-butyl bromide with a substitution of *t*-butyl bromide, we get the reaction shown in the margin. It turns out that, kinetically, this reaction is first order: its rate depends only on the concentration of *tert*-BuBr—it doesn't matter how much hydroxide you add: the rate equation is simply



$$\text{rate} = k_1[t\text{-BuBr}]$$

The reason for this is that the reaction happens in two steps: first the bromide leaves, to generate a carbocation, and only then does the hydroxide ion move in to attack, forming the alcohol.

the S_N1 mechanism: reaction of $t\text{-BuBr}$ with hydroxide ion



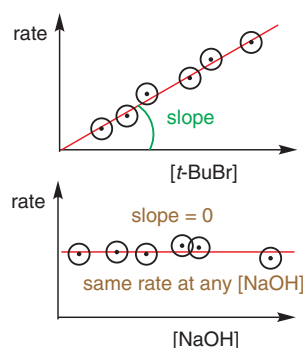
stage 1: formation of the carbocation



stage 2: reaction of the carbocation

In the S_N1 mechanism, the formation of the cation is the rate-determining step. This makes good sense: a carbocation is an unstable species and so it will be formed slowly from a stable neutral organic molecule. But once formed, being very reactive, all its reactions will be fast, regardless of the nucleophile. The rate of disappearance of $t\text{-BuBr}$ is therefore simply the rate of the slow first step: the hydroxide nucleophile is not involved in this step and therefore does not appear in the rate equation and hence cannot affect the rate. If this is not clear to you, think of a crowd of people trying to leave a railway station or a football match through some turnstiles. It doesn't matter how fast they walk, run, or are driven away in taxis afterwards, it is only the rate of struggling through the turnstiles that determines how fast the station or stadium empties.

Once again, this rate equation is useful because we can determine whether a reaction is S_N1 or S_N2 . We can plot the same graphs as we plotted before. If the reaction is S_N2 , the graphs look like those we have just seen. But if it is S_N1 , the graphs in the margin show what happens when we vary $[t\text{-BuBr}]$ at constant $[\text{NaOH}]$ and then vary $[\text{NaOH}]$ at constant $[t\text{-BuBr}]$.



The slope of the first graph is simply the first-order rate constant because $\text{rate} = k_1[t\text{-BuBr}]$. But the slope of the second graph is zero. The rate-determining step does not involve NaOH so adding more of it does not speed up the reaction. The reaction shows first-order kinetics (the rate is proportional to one concentration only) and the mechanism is called S_N1 , that is, Substitution, Nucleophilic, 1st order.

This observation is very significant. The fact that the nucleophile does not appear in the rate equation means that not only does its *concentration* not matter—its *reactivity* doesn't matter either! We are wasting our time opening a tub of NaOH to add to this reaction—water will do just as well. All the oxygen nucleophiles in the table above react at the *same* rate with $t\text{-BuBr}$ although they react at very different rates with MeI . Indeed, S_N1 substitution reactions are generally best done with weaker, non-basic nucleophiles to avoid the competing elimination reactions discussed in Chapter 17.

● **The rate of an S_N1 reaction depends on:**

- the carbon skeleton
- the leaving group

along with the usual factors of temperature and solvent.

But NOT the nucleophile.

How can we decide which mechanism (S_N1 or S_N2) will apply to a given organic compound?

So, substitution reactions at saturated C go via one of two alternative mechanisms, each with a very different dependence on the nature of the nucleophile. It's important to be able to predict which mechanism is likely to apply to any reaction, and rather than doing the kinetic experiments to find out, we can give you a few simple pointers to predict which will operate

in which case. The factors that affect the mechanism of the reaction also help to explain why that mechanism operates.

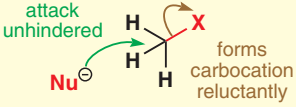
The most important factor is **the structure of the carbon skeleton**. A helpful generalization is that compounds that can form relatively stable carbocations generally do so and react by the S_N1 mechanism, while the others have no choice but to react by the S_N2 mechanism. As you will see in a moment, the most stable carbocations are the ones that have the most substituents, so the more carbon substituents at the reaction centre, the more likely the compound is to react by the S_N1 mechanism.

As it happens, the structural factors that make cations stable usually also lead to slower S_N2 reactions. Heavily substituted compounds are good in S_N1 reactions, but bad in an S_N2 reaction because the nucleophile would have to squeeze its way into the reaction centre past the substituents. It is better for an S_N2 reaction if there are only hydrogen atoms at the reaction centre—methyl groups react fastest by the S_N2 mechanism. The effects of the simplest structural variations are summarized in the table below (where R is a simple alkyl group like methyl or ethyl).

● S_N1 or S_N2 ?

Simple structures and choice of S_N1 or S_N2 mechanism

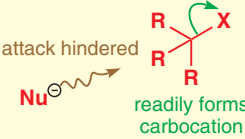
Structure type	$\text{Me}-\text{X}$			
S_N1 reaction?	no	no	moderate	excellent
S_N2 reaction?	good	good	moderate	no



attack unhindered

Nu^-

forms carbocation reluctantly



attack hindered

Nu^-

readily forms carbocation

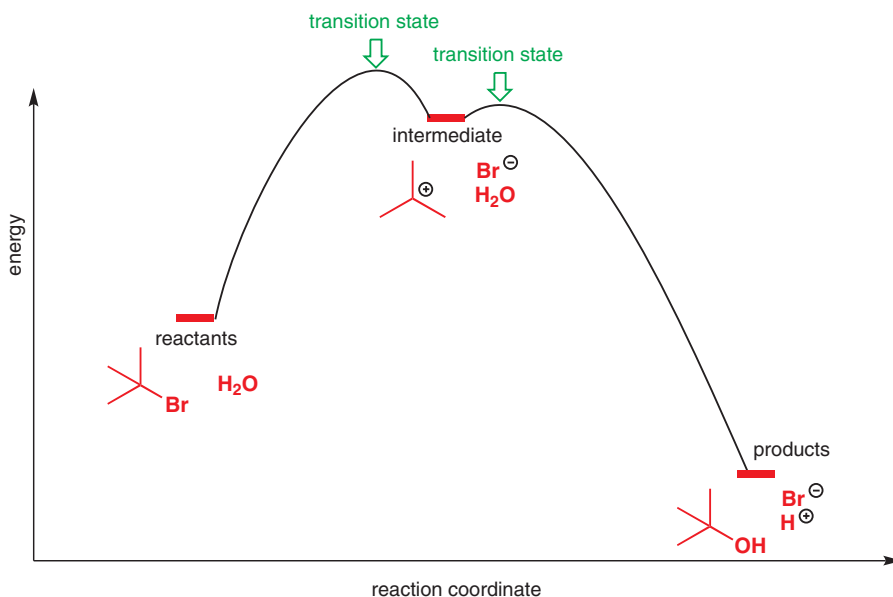
The only doubtful case is the secondary alkyl derivative, which can react by either mechanism, although it is not very good at either. The first question you should ask when faced with a new nucleophilic substitution is this ‘Is the carbon electrophile methyl, primary, secondary, or tertiary?’ This will start you off on the right foot, which is why we introduced these important structural terms in Chapter 2.

Later in this chapter we will look in more detail at the differences between the two mechanisms and the structures that favour each, but all of what we say will build on the table above.

A closer look at the S_N1 reaction

In our discussion of the S_N1 reaction above, we proposed the *t*-butyl carbocation as a reasonable intermediate formed by loss of bromide from *t*-butyl bromide. We now need to explain the evidence we have that carbocations can indeed exist, and the reasons why the *t*-butyl carbocation is much more stable than, for example, the *n*-butyl cation.

In Chapter 12 we introduced the idea of using a reaction energy profile diagram to follow the progress of a reaction from starting materials to products, via transition states and any intermediates. The energy profile diagram for the S_N1 reaction between *t*-butyl bromide and water looks something like this:



The carbocation is shown as an intermediate—a species with a finite (if short) lifetime for reasons we shall describe shortly. And because we know that the first step, the formation of the carbocation, is slow, that must be the step with the higher energy transition state. The energy of that transition state, which determines the overall rate of the reaction, is closely linked to the stability of the carbocation intermediate, and it is for this reason that the most important factor in determining the efficiency of an S_N1 reaction is the stability or otherwise of any carbocation that might be formed as an intermediate.

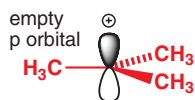
Shape and stability of carbocations

We discussed the planar shape of the methyl cation in Chapter 4 (p. 103), and the *tert*-butyl cation is similar in structure: the electron-deficient central carbon atom has only six electrons, which it uses to form three σ bonds, and therefore also carries an empty p orbital. Any carbocation will have a planar carbon atom with an empty p orbital. Think of it this way: only filled orbitals contribute to the energy of a molecule, so if you have to have an unfilled orbital (which a carbocation always does) it is best to make that unfilled orbital as high in energy as possible to keep the filled orbitals low in energy. p orbitals are higher in energy than s orbitals (or hybrid sp , sp^2 , or sp^3 orbitals for that matter) so the carbocation always keeps the p orbital empty.

Carbocation stability

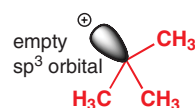
The *t*-butyl carbocation is *relatively* stable as far as carbocations go, but you would not be able to keep it in a bottle on the shelf! The concept of more and less stable carbocations is important in understanding the S_N1 reaction, but it is important to realize that these terms are all relative: even ‘stable’ carbocations are highly reactive electron-deficient species.

correct planar structure
for the *t*-butyl cation



less repulsion
between
bonding pairs of
electrons

incorrect tetrahedral structure
for the *t*-butyl cation

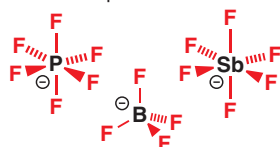


more repulsion
between
bonding pairs of
electrons

We know that the *t*-butyl cation is stable enough to observe because of the work of George Olah, who won the Nobel Prize for Chemistry in 1994. The challenge is that carbocations are very reactive electrophiles, so Olah's idea was to have a solution containing no nucleophiles. Any cation must have an anion to balance the charge, so the important advance was to find anions, consisting of a negatively charged atom surrounded by tightly held halogen atoms, which are just too stable to be nucleophilic. Examples include BF_4^- , PF_6^- , and SbF_6^- . The first is small and tetrahedral and the others are larger and octahedral.

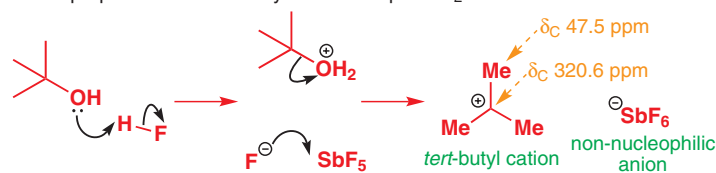
In these anions, the negative charge does not correspond to a lone pair of electrons (they are like BH_4^- in this respect) and there is no orbital high enough in energy to act as a nucleophile. By using a non-nucleophilic solvent, liquid SO_2 , at low temperature, Olah was able to turn alcohols into carbocations with these counterions. This is what happens when *tert*-butanol is treated with SbF_5 and HF in liquid SO_2 . The acid protonates the hydroxyl group, allowing it to

non-nucleophilic anions



leave as water, while the SbF₅ grabs the fluoride ion, preventing it from acting as a nucleophile. The cation is left high and dry.

Olah's preparation of the *t*-butyl cation in liquid SO₂



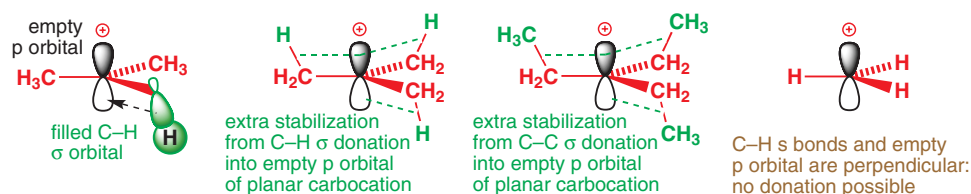
The proton NMR of this cation showed just one signal for the three methyl groups at 4.15 ppm, quite far downfield for C–Me groups. The ¹³C spectrum also showed downfield Me groups at 47.5 ppm, but the key evidence that the cation was formed was the shift of the central carbon atom, which came at an amazing 320.6 ppm, way downfield from anything you have met before. This carbon is very deshielded—it is positively charged and extremely electron deficient.

From Olah's work we know what the *t*-butyl cation looks like by NMR, so can we use NMR to try to detect it as an intermediate in substitution reactions? If we mix *t*-BuBr and NaOH in an NMR tube and let the reaction run inside the NMR machine, we see no signals belonging to the cation. But this proves nothing. We would not expect a reactive intermediate to be present in any significant concentration. There is a simple reason for this. If the cation is unstable, it will react very quickly with any nucleophile around and there will never be any appreciable amount of cation in solution. Its rate of formation will be much slower than its rate of reaction.

Alkyl substituents stabilize a carbocation

Olah found that he could measure the spectrum of the *tert*-butyl cation, but he was never able to observe the methyl cation in solution. Why do those extra substituents stabilize the cationic centre?

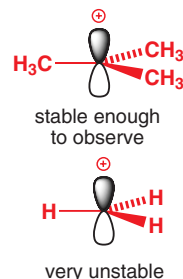
Any charged organic intermediate is inherently unstable because of the charge. A carbocation can be formed only if it has some extra stabilization, and extra stabilization can come to the planar carbocation structure from weak donation of σ bond electrons into the empty p orbital of the cation. In the *t*-butyl cation, three of these donations occur at any one time: it doesn't matter if the C–H bonds point up or down; one C–H bond of each methyl group must be parallel to one lobe of the empty p orbital at any one time. The first diagram shows one overlap in orbital terms and the second and third diagrams, three as dotted lines.



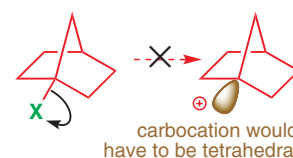
There is nothing special about the C–H bond donating electrons into an empty orbital: a C–C bond is just as good and some bonds are better (C–Si, for example). But there must be a bond of some sort—a hydrogen atom by itself has no lone pairs and no σ bonds so it cannot stabilize a cation.

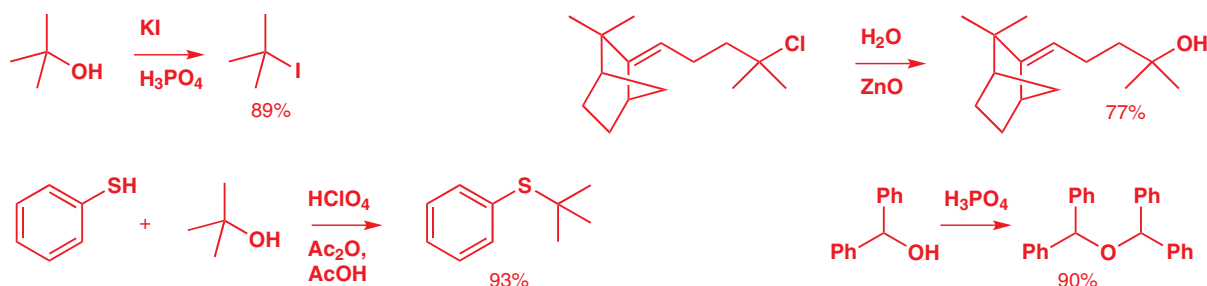
Planarity is so important to the structure of a carbocation that if a tertiary cation cannot become planar, it is not formed. A classic case is the structure in the margin, which does not react with nucleophiles either by S_N1 or by S_N2. It does not react by S_N1 because the cation cannot become planar, nor by S_N2 because the nucleophile cannot approach the carbon atom from the right direction.

In general, though, simple tertiary structures undergo efficient S_N1 substitution reactions. With good leaving groups such as halides, substitutions can be done under neutral conditions; with less good leaving groups such as alcohols or ethers, acid catalysis is required. The following group of reactions give an idea of the types of S_N1 reactions that work well.



Interactive display of stability and structure of carbocations





An adjacent C=C π system stabilizes a carbocation: allylic and benzylic carbocations

➡ We discussed conjugation in allyl cations in Chapter 7.

Tertiary carbocations are more stable than primary ones, but powerful stabilization is also provided when there is genuine conjugation between the empty p orbital and adjacent π or lone pair electrons. The allyl cation has a filled (bonding) orbital containing two electrons delocalized over all three atoms and an important empty orbital with coefficients on the end atoms only. It's this orbital that is attacked by nucleophiles. The curly arrow picture tells us the same thing.

the allyl cation

curly arrows

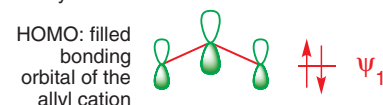


molecular orbitals

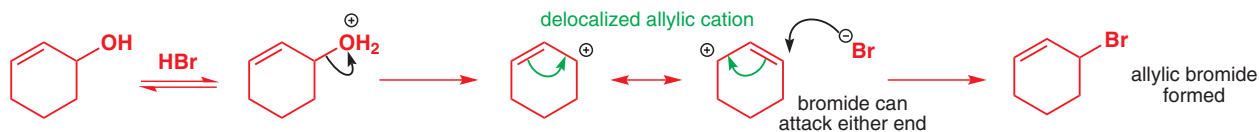
LUMO: empty non-bonding orbital of the allyl cation



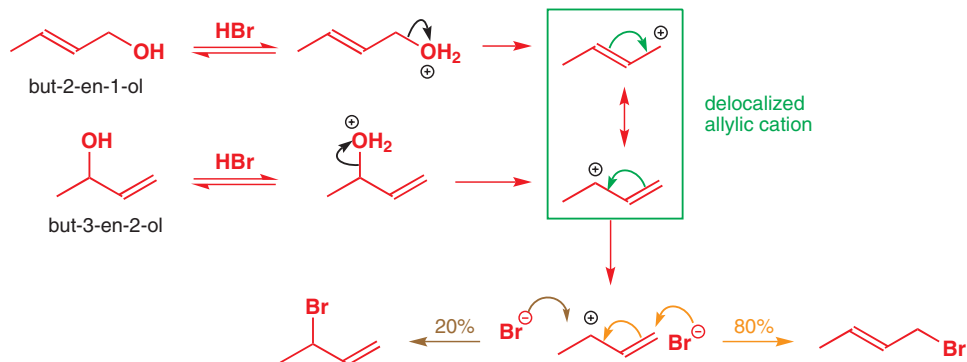
HOMO: filled bonding orbital of the allyl cation



Allylic electrophiles react well by the S_N1 mechanism because the allyl cation is relatively stable. Here's an example of a reaction working in the opposite direction from most of those you have seen so far—we start with the alcohol and form the bromide. Treatment of cyclohexenol with HBr gives the corresponding allylic bromide.

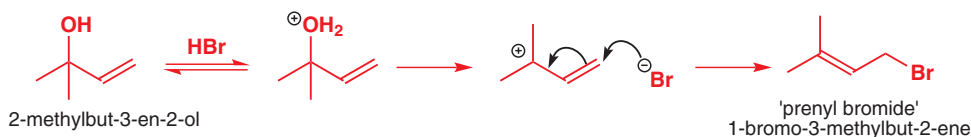


In this case, only one compound is formed because attack at either end of the allylic cation gives the same product. But when the allylic cation is unsymmetrical this can be a nuisance as a mixture of products may be formed. It doesn't matter which of these two butenols you treat with HBr, you get the same delocalized allylic cation.

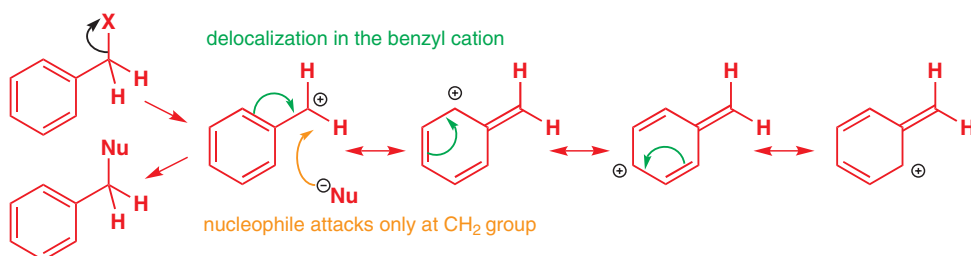


When this cation reacts with Br^- , about 80% goes to one end and 20% to the other, giving a mixture of butenyl bromides. This **regioselectivity** (where the nucleophile attacks) is determined by steric hindrance: attack is faster at the less hindered end of the allylic system.

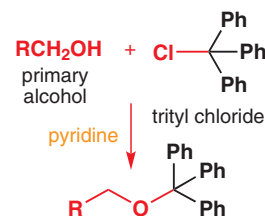
Sometimes this ambiguity is useful. The tertiary allylic alcohol 2-methylbut-3-en-2-ol is easy to prepare and reacts well by the S_N1 mechanism because it can form a stable carbocation that is both tertiary and allylic. The allylic carbocation intermediate is unsymmetrical and reacts only at the less substituted end to give 'prenyl bromide'.



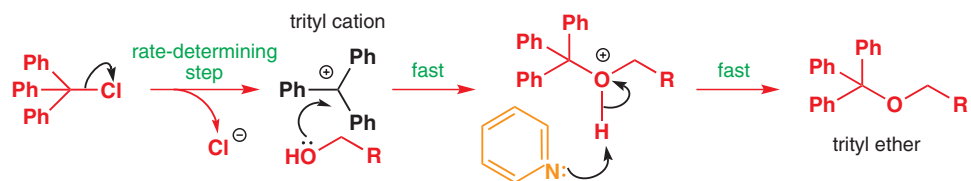
The benzyl cation is about as stable as the allyl cation but lacks its ambiguity of reaction. Although the positive charge is delocalized around the benzene ring, to three positions in particular, the benzyl cation always reacts on the side chain so that aromaticity is preserved.



An exceptionally stable cation is formed when three benzene rings can help to stabilize the same positive charge. The result is the triphenylmethyl cation or, for short, the trityl cation. Trityl chloride is used to form an ether with a primary alcohol group by an S_N1 reaction. You will notice that pyridine is used as solvent for the reaction. Pyridine (a weak base: the pK_a of its conjugate acid is 5.5—see Chapter 8) is not strong enough to remove the proton from the primary alcohol (pK_a about 15), and there would be no point in using a base strong enough to make RCH_2O^- as the nucleophile makes no difference to an S_N1 reaction. Instead the TrCl ionizes first to trityl cation, which now captures the primary alcohol and finally pyridine is able to remove the proton from the oxonium ion. Pyridine does not catalyse the reaction; it just stops it becoming too acidic by removing the HCl formed. Pyridine is also a convenient polar organic solvent for ionic reactions.

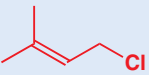


■ The symbol Tr refers to the group Ph_3C .



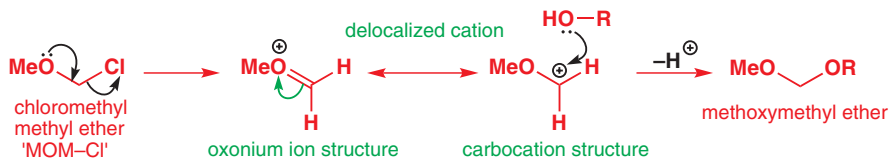
The table below shows the rates of solvolysis (i.e. a reaction in which the solvent acts as the nucleophile) in 50% aqueous ethanol for substituted allylic chlorides compared with benzylic chlorides and simple alkyl chlorides. The values give you an idea of the relative reactivity towards substitution of the different classes of compound. These rates are mostly S_N1 , but there will be some S_N2 reactivity with the primary compounds.

Rates of solvolysis of alkyl chlorides in 50% aqueous ethanol at 44.6 °C

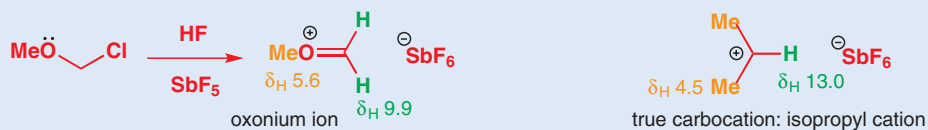
Compound	Relative rate	Comments
	0.07	primary chloride: probably all S _N 2
	0.12	secondary chloride: can do S _N 1 but not very well
	2100	tertiary chloride: very good at S _N 1
	1.0	primary but allylic: S _N 1 all right
	91	allylic cation is secondary at one end
	130000	allylic cation is tertiary at one end: compare with 2100 for simple tertiary
	7700	primary but allylic and benzylic

Carbocations are stabilized by an adjacent lone pair

The alkyl chloride known as methyl chloromethyl ether, MeOCH_2Cl , reacts very well with alcohols to form ethers. Being a primary alkyl chloride, you might think that its reactions would follow an $\text{S}_{\text{N}}2$ mechanism, but in fact it has characteristic $\text{S}_{\text{N}}1$ reactivity. As usual, the reason for its preference for the $\text{S}_{\text{N}}1$ mechanism is its ability to form a stabilized carbocation. Loss of the chloride ion is assisted by the adjacent lone pair, and we can draw the resulting cation either as an oxonium ion or as a carbocation.

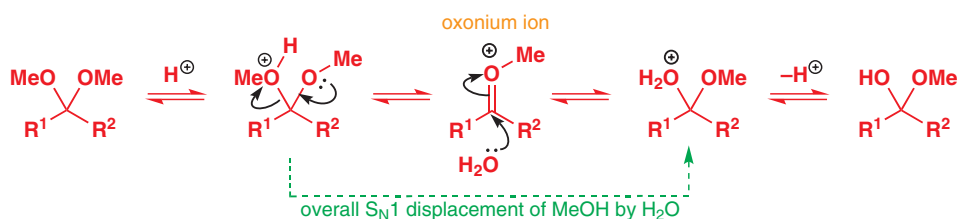


The methoxymethyl cation



Olah has used the methods described above to make the methoxymethyl cation in solution. Although this cation can be drawn either as an oxonium ion or as a primary carbocation, the oxonium ion structure is the more realistic. The proton NMR spectrum of the cation compared with that of the isopropyl cation (this is the best comparison we can make) shows that the protons on the CH_2 group resonate at 9.9 ppm instead of at the 13.0 ppm of the true carbocation.

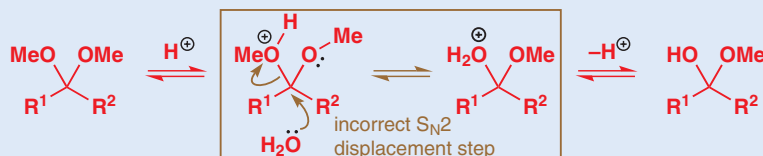
If you think back to Chapter 11, you will recall that the first step in the hydrolysis of an acetal is a similar reaction, with one alkoxy group replaced by water to give a hemiacetal. We considered the mechanism for this reaction in Chapter 11 but did not then concern ourselves with a label for the first step. It is in effect an $\text{S}_{\text{N}}1$ substitution reaction: the decomposition of the protonated acetal to give an oxonium ion. If you compare this step with the reaction of the chloroether we have just described you will see that they are very similar in mechanism.



Interactive mechanism for acetal hydrolysis

A common mistake

Don't be tempted to shortcut this mechanism by drawing the displacement of the first molecule of methanol by water as an S_N2 reaction.



An S_N2 mechanism is unlikely at such a crowded carbon atom. However, the main reason why the S_N2 mechanism is wrong is that the S_N1 mechanism is so very efficient, with a neighbouring MeO group whose lone pair can stabilize the carbocation intermediate. The S_N2 mechanism doesn't get a chance.

This mechanism for the S_N1 replacement of one electronegative group at a carbon atom by a nucleophile where there is another electronegative group at the same carbon atom is very general. You should look for it whenever there are two atoms such as O, N, S, Cl, or Br joined to the same carbon atom. The better leaving groups (such as the halogens) need no acid catalyst but the less good ones (N, O, S) usually need acid.



➡ Think back to the formation and reactions of iminium ions in Chapter 11 for further examples.

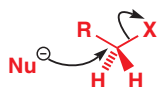
We now have in the box below a complete list of the sorts of structures that normally react by the S_N1 mechanism rather than by the S_N2 mechanism.

Stable carbocations as intermediates in S_N1 reactions

Type of cations	Example 1	Example 2
simple alkyl	tertiary (good) <i>t</i> -butyl cation $\text{Me}_3\text{C}^+ = \text{Me}-\text{C}^+(\text{Me})_2$	secondary (not so good) <i>i</i> -propyl cation $\text{Me}_2\text{CH}^+ = \text{H}-\text{C}^+(\text{Me})_2$
conjugated	allylic 	benzylic
heteroatom-stabilized	oxygen-stabilized (oxonium ions) 	nitrogen-stabilized (iminium ions)

A closer look at the S_N2 reaction

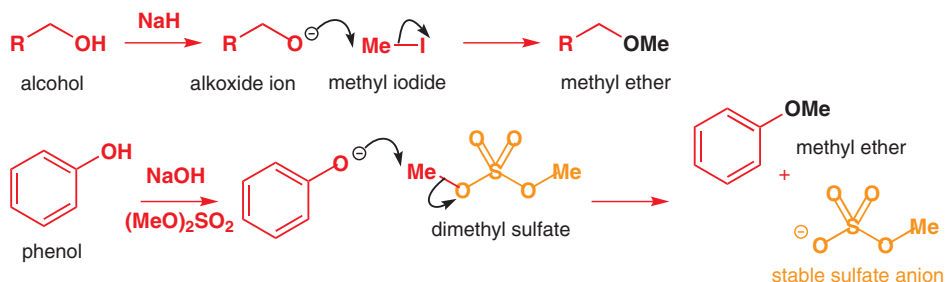
■ Notice that we said *simple* alkyl groups: of course, primary allylic, benzylic, and RO or R₂N substituted primary derivatives may react by S_N1!



uncluttered approach of nucleophile in S_N2 reactions of methyl compounds (R=H) and primary alkyl compounds (R=alkyl)

Among simple alkyl groups, methyl and primary alkyl groups always react by the S_N2 mechanism and never by S_N1. This is partly because the cations are unstable and partly because the nucleophile can push its way in easily past the hydrogen atoms.

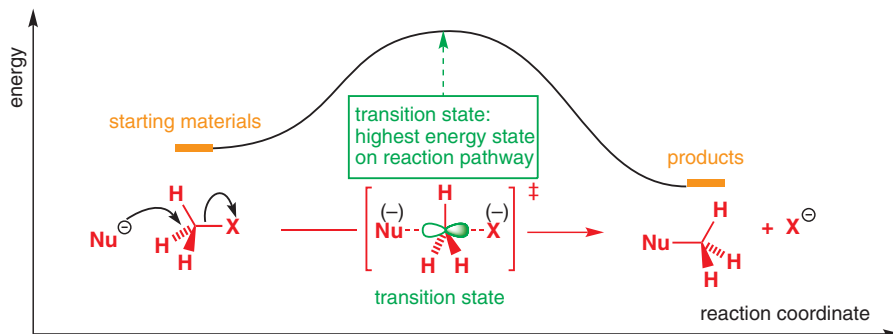
A common way to make ethers is to treat an alkoxide anion with an alkyl halide. If the alkyl halide is a methyl compound, we can be sure that the reaction will go by the S_N2 mechanism. A strong base, here NaH, will be needed to form the alkoxide ion, since alcohols are weak acids (pK_a about 16). Methyl iodide is a suitable electrophile.



With the more acidic phenols (pK_a about 10), NaOH is a strong enough base and dimethyl sulfate, the dimethyl ester of sulfuric acid, is often used as the electrophile. It is worth using a strong base to make the alcohol into a better nucleophile because as we discussed on p. 331 the rate equation for an S_N2 reaction tells us that the strength and concentration of the nucleophile affects the rate of the reaction.

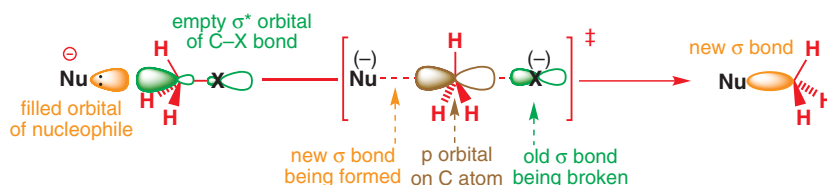
The transition state for an S_N2 reaction

Another way to put this would be to say that the nucleophile, the methyl group, and the leaving group are all present in the transition state for the reaction. The transition state is the highest energy point on the reaction pathway. In the case of an S_N2 reaction it will be the point where the new bond from the nucleophile is partly formed while the old bond to the leaving group is not yet completely broken. It will look something like this:



The dashed bonds in the transition state indicate partial bonds (the C–Nu bond is partly formed and the C–X bond partly broken) and the charges in brackets indicate substantial partial charges (about half a minus charge each in this case). Transition states are often shown in square brackets and marked with the symbol ‡.

Another way to look at this situation is to consider the orbitals. The nucleophile must have lone-pair electrons, which will interact with the σ* orbital of the C–X bond.



➡ We introduced the terms *transition state* and *intermediate* in Chapter 12.

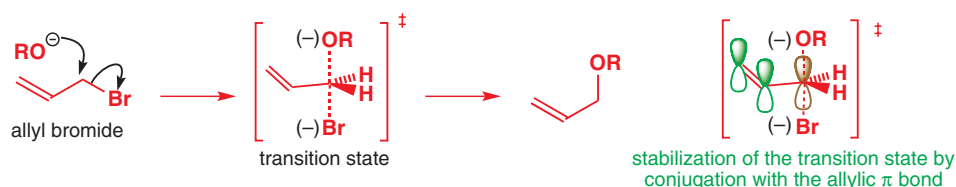
🖱 Interactive mechanism for simple S_N2

In the transition state the carbon atom in the middle has a p orbital that shares one pair of electrons between the old and the new bonds. Both these pictures suggest that the transition state for an S_N2 reaction has a more or less planar carbon atom at the centre with the nucleophile and the leaving group arranged at 180° to each other. This picture can help us explain two important observations concerning the S_N2 reaction—firstly the types of structures that react efficiently, and secondly the stereochemistry of the reaction.

Adjacent C=C or C=O π systems increase the rate of S_N2 reactions

We have already established that methyl and primary alkyl compounds react well by the S_N2 mechanism, while secondary alkyl compounds undergo S_N2 reactions only reluctantly. But there are other important structural features that also encourage the S_N2 mechanism. Two of these, allyl and benzyl groups, also encourage the S_N2 mechanism.

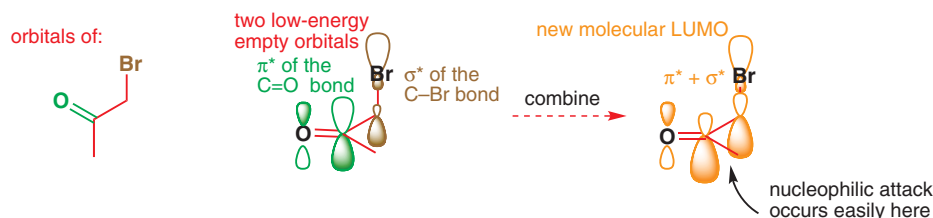
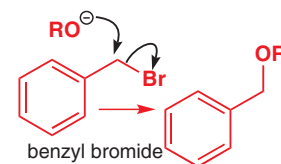
Allyl bromide reacts well with alkoxides to make ethers, and shown below is the typical S_N2 mechanism for this reaction. Also shown is the transition state for this reaction. Allyl compounds react rapidly by the S_N2 mechanism because the π system of the adjacent double bond can stabilize the transition state by conjugation. The p orbital at the reaction centre (shown in brown, and corresponding to the brown orbital in the diagram on p. 340) has to make two partial bonds with only two electrons—it is electron deficient, and so any additional electron density it can gather from an adjacent π system will stabilize the transition state and increase the rate of the reaction.



Interactive S_N2 mechanism at allylic and benzylic centres

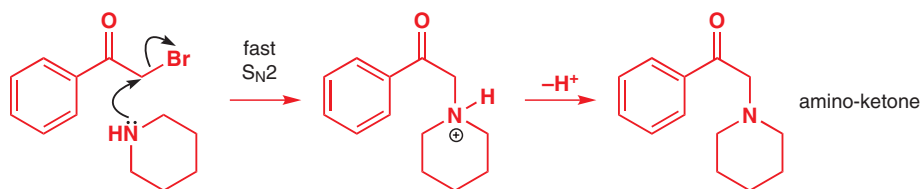
The benzyl group acts in much the same way using the π system of the benzene ring for conjugation with the p orbital in the transition state. Benzyl bromide reacts very well with alkoxides to make benzyl ethers.

Among the fastest of all S_N2 reactions are those where the leaving group is adjacent to a carbonyl group. With α -bromo carbonyl compounds, two neighbouring carbon atoms are both powerfully electrophilic sites. Each has a low-energy empty orbital— π^* from C=O and σ^* from C—Br (this is what makes them electrophilic)—and these can combine to form a new LUMO ($\pi^* + \sigma^*$) lower in energy than either. Nucleophilic attack will occur easily where this new orbital has its largest coefficient, shown in orange on the diagram.



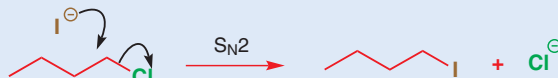
The effect of this interaction between antibonding orbitals is that each group becomes more electrophilic because of the presence of the other—the C=O group makes the C—Br bond more reactive and the Br makes the C=O group more reactive. In fact, it may well be that the nucleophile will attack the carbonyl group, but this will be reversible whereas displacement of bromide is irreversible.

There are many examples of this type of reaction. Reactions with amines go well and the aminoketone products are widely used in the synthesis of drugs.



Quantifying structural effects on S_N2 reactions

Some actual data may help at this point. The rates of reaction of the following alkyl chlorides with KI in acetone at 50 °C broadly illustrate the patterns of S_N2 reactivity we have just analysed. These are relative rates with respect to n -BuCl as a 'typical primary halide'. You should not take too much notice of precise figures but rather observe the trends and notice that the variations are quite large—the full range from 0.02 to 100,000 is eight powers of ten.



Relative rates of substitution reactions of alkyl chlorides with the iodide ion

Alkyl chloride	Relative rate	Comments
$\text{Me}-\text{Cl}$	200	least hindered alkyl chloride
	0.02	secondary alkyl chloride; slow because of steric hindrance
	79	allyl chloride accelerated by π conjugation in transition state
	200	benzyl chloride a bit more reactive than allyl: benzene ring slightly better at π conjugation than isolated double bond
$\text{Me}-\text{O}-\text{CH}_2-\text{Cl}$	920	conjugation with oxygen lone pair accelerates reaction (this is an S_N1 reaction)
	100,000	conjugation with carbonyl group much more effective than with simple alkene or benzene ring; these α -halo carbonyl compounds are the most reactive of all

Contrasts between S_N1 and S_N2

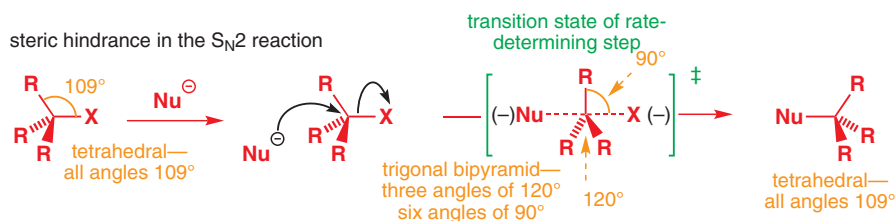
You have now met the key features of both important mechanisms for substitution. You should at this stage in the chapter have a grasp of the kinetics, the nature of the intermediates and transition states, and the simple steric and electronic factors that control reactivity in S_N1 and S_N2 reaction pathways.

We are now going to look in more detail at some other aspects where there are significant contrasts between the mechanisms, either because they lead to different outcomes or because they lead to a change in reactivity towards one or the other of the two pathways.

A closer look at steric effects

We have already pointed out that having more alkyl substituents at the reaction centre makes a compound more likely to react by S_N1 than by S_N2 for two reasons: firstly they make a carbocation more stable, so favouring S_N1 , and secondly they make it hard for a nucleophile to get close to the reaction centre in the rate-determining step, disavouring S_N2 . Let's look in more detail at the transition state for the slow steps of the two reactions and see how steric hindrance affects both.

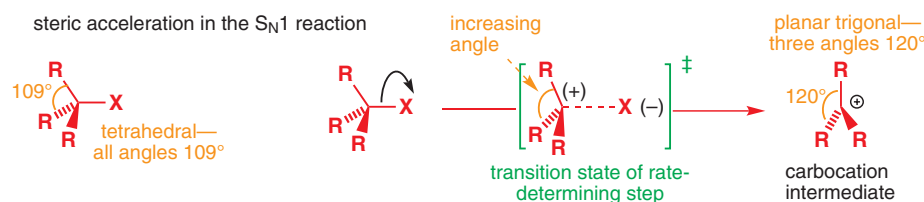
In the approach to the S_N2 transition state, the carbon atom under attack gathers in another substituent and becomes (transiently) five-coordinate. The angles between the substituents decrease from tetrahedral to about 90°.



In the starting material there are four angles of about 109°. In the transition state (enclosed in square brackets and marked ‡ as usual) there are three angles of 120° and six angles of 90°, a significant increase in crowding. The larger the substituents R, the more serious this is, and the greater the increase in energy of the transition state. We can easily see the effects of steric hindrance if we compare these three structural types:

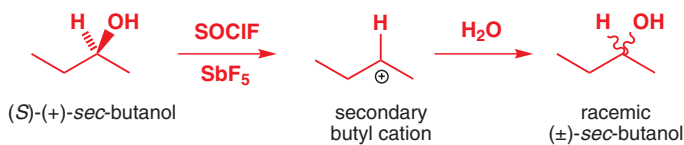
- methyl: CH₃–X: very fast S_N2 reaction
- primary alkyl: RCH₂–X: fast S_N2 reaction
- secondary alkyl: R₂CH–X: slow S_N2 reaction.

The opposite is true of the S_N1 reaction. The rate-determining step is simply the loss of the leaving group, and the transition state for this step will look something like the structure shown below—with a longer, weaker, and more polarized C–X bond than the starting material. The starting material is again tetrahedral (four angles of about 109°) and in the intermediate cation there are just three angles of 120°—fewer and less serious interactions. The transition state will be on the way towards the cation, and because the R groups are further apart in the transition state than in the starting material, large R groups will actually *decrease* the energy of the transition state relative to the starting material. S_N1 reactions are therefore accelerated by alkyl substituents both for this reason and because they stabilize the cation.



Stereochemistry and substitution

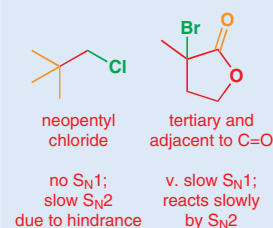
Look back at the scheme we showed you for the S_N2 reaction on p. 340. It shows the nucleophile attacking the carbon atom on the opposite side from the leaving group. Look carefully at the carbon atom it is attacking and you see that its substituents end up turning inside out as the reaction goes along, just like an umbrella in a high wind. If the carbon atom under attack is a stereogenic centre (Chapter 14), the result will be inversion of configuration. Something very different happens in the S_N1 reaction, and we will now illustrate the difference with a simple sequence of reactions.



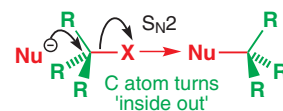
Starting with the optically active secondary alcohol *sec*-butanol (or butan-2-ol, but we want to emphasize that it is *secondary*), the secondary cation can be made by the method described on p. 338. Quenching this cation with water regenerates the alcohol but without any optical

You may hear it said that *tert*-alkyl compounds do not react by the S_N2 mechanism because the steric hindrance would be too great. In fact *t*-alkyl compounds react so fast by the S_N1 mechanism that the S_N2 mechanism wouldn't get a chance *even if it went as fast as it goes with methyl compounds*. You can see the figures that show this in the box on p. 338. Some tertiary alkyl compounds will react by slow S_N2 if, for example, S_N1 is prevented by an adjacent electron-withdrawing carbonyl group.

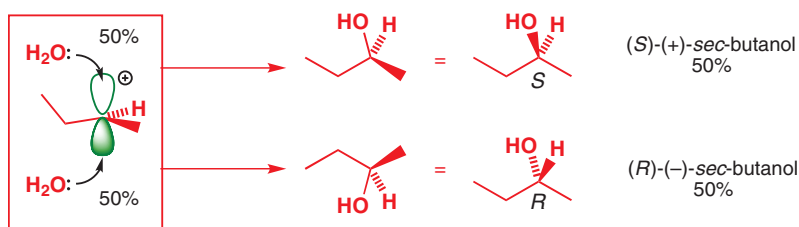
Likewise, primary alkyl halides that experience steric hindrance for other reasons react neither by S_N1 (because they are primary) nor S_N2 (because they are hindered)—a notorious example is the unreactivity of 'neopentyl halides'.



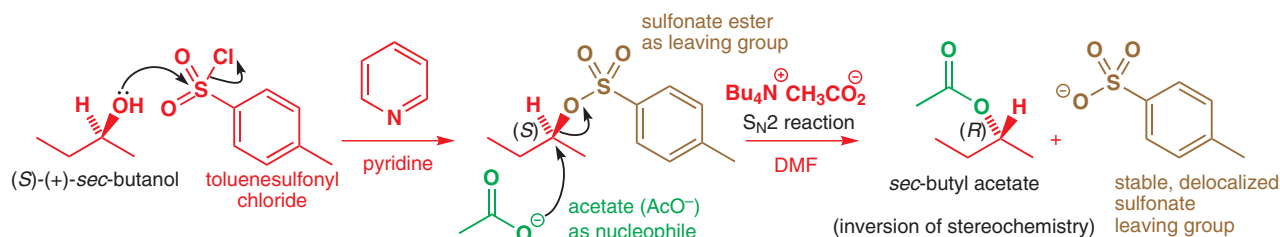
Notice how the transition state for the S_N1 reaction is really very close in structure to the carbocation, and you can see that they also lie very close to one another in the energy profile diagram on p. 334. When we say that the rate of an S_N1 reaction is increased by stabilization of the carbocation, what we really mean of course is that the rate is increased by stabilization of the *transition state leading to* the carbocation. However, they are so similar in structure that you can assume that steric and electronic effects on the carbocation will be very similar to those on the carbocation.



activity. Water must attack the two faces of the planar cation with exactly equal probability: the product is an exactly 50:50 mixture of (*S*)-butanol and (*R*)-butanol. It is *racemic*.

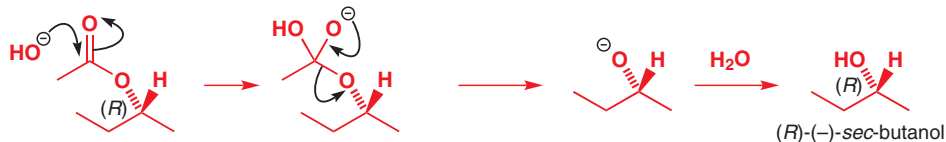


Alternatively, we can first make the hydroxyl group into a good enough leaving group to take part in an $\text{S}_{\text{N}}2$ reaction. The leaving group we shall use, a sulfonate ester, will be introduced to you in a few pages' time, but for now you just need to accept that nucleophilic attack of the OH group on a sulfonyl chloride in pyridine solution gives the sulfonate ester shown below in orange: no bonds have been formed or broken at the chiral carbon atom, which still has (*S*) stereochemistry.



Now we can carry out an $\text{S}_{\text{N}}2$ reaction on the sulfonate with an acetate anion. A tetra-alkyl ammonium salt is used in the solvent DMF to avoid solvating the acetate, making it as powerful a nucleophile as possible and getting a clean $\text{S}_{\text{N}}2$ reaction. This is the key step and we don't want any doubt about the outcome. The sulfonate is an excellent leaving group—the charge is delocalized across all three oxygen atoms.

The product *sec*-butyl acetate is optically active and we can measure its optical rotation. But this tells us nothing. Unless we know the true rotation for pure *sec*-butyl acetate, we don't yet know whether it is optically pure nor even whether it really is inverted. We expect it to have (*R*) stereochemistry, but we can easily find out for sure. All we have to do is to hydrolyse the ester and get the original alcohol back again. We know the true rotation of the alcohol—it was our starting material—and we know that ester hydrolysis (Chapter 10) proceeds by attack at the carbonyl carbon—it can't affect the stereochemistry of the chiral centre.



Now we really know where we are. This new sample of *sec*-butanol has the same rotation as the original sample, *but with the opposite sign*. It is (-)-(*R*)-*sec*-butanol. It is optically pure and inverted. Somewhere in this sequence there has been an inversion, and we know it wasn't in the formation of the sulfonate or the hydrolysis of the acetate as no bonds are formed or broken at the stereogenic centre in these steps. It must have been in the $\text{S}_{\text{N}}2$ reaction itself.

- An $\text{S}_{\text{N}}2$ reaction goes with inversion of configuration at the carbon atom under attack but an $\text{S}_{\text{N}}1$ reaction generally goes with racemization.

The effect of solvent

Why was the $\text{S}_{\text{N}}2$ reaction we have just shown you carried out in DMF? You will generally find $\text{S}_{\text{N}}2$ reactions are carried out in aprotic, and often less polar, solvents. $\text{S}_{\text{N}}1$ reactions are

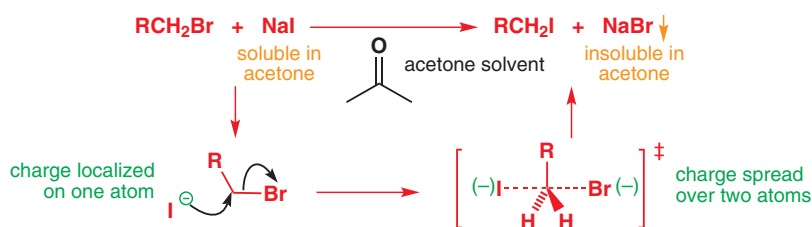
➡ Optical rotation is described on p. 309.

➡ The different types of solvents were discussed in Chapter 12.

typically carried out in polar, protic solvents. A common solvent for an S_N2 reaction is acetone—just polar enough to dissolve the ionic reagents, but not as polar as, say, acetic acid, a common solvent for the S_N1 reaction.

It is fairly obvious why the S_N1 reaction needs a polar solvent: the rate-determining step involves the formation of ions (usually a negatively charged leaving group and a positively charged carbocation) and the rate of this process will be increased by a polar solvent that can solvate these ions. More precisely, the transition state is more polar than the starting materials (note the charges in brackets in the scheme above) and so is stabilized by the polar solvent. Hence solvents like water or carboxylic acids (RCO₂H) are ideal.

It is less obvious why a less polar solvent is better for the S_N2 reaction. The most common S_N2 reactions use an anion as the nucleophile. The transition state is then less polar than the localized anion as the charge is spread between two atoms. Here's an example: the formation of an alkyl iodide from an alkyl bromide. Acetone fails to solvate the iodide well, making it more reactive; the transition state is less in need of solvation, so overall the reaction is faster.

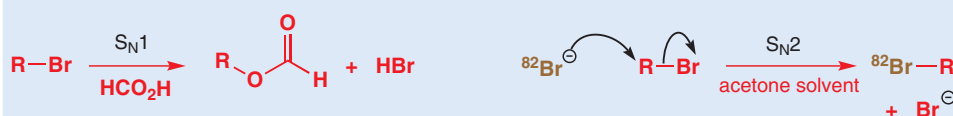


■ Acetone also assists this reaction because it dissolves sodium iodide but not the sodium bromide product, which precipitates from solution and prevents bromide acting as a competing nucleophile.

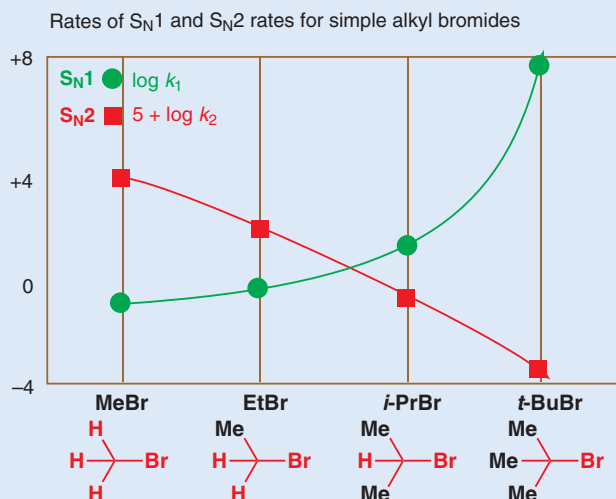
DMF and DMSO, the polar aprotic solvents we discussed in Chapter 12 (p. 255) are also good solvents for S_N2 reactions because they dissolve ionic compounds well but fail to solvate anions well, making them more reactive. The choice of Bu₄N⁺—a large, non-coordinating cation—as the counterion for the reaction on p. 344 was also made with this in mind.

Quantifying the rates of S_N1 and S_N2 reactions

The data below illustrate the effect of structure on the rates of S_N1 and S_N2 reactions. The green curve on the graph shows the rates (k_1) of an S_N1 reaction: the conversion of alkyl bromides to alkyl formate esters in formic acid at 100 °C. Formic acid is a polar solvent and a weak nucleophile: perfect for an S_N1 reaction. The red curve shows the rates of displacement of Br[−] by radioactive ⁸²Br[−] in acetone at 25 °C. Acetone solvent and the good nucleophile Br[−] favour S_N2. The rates (k_2) are multiplied by 10⁵ to bring both curves onto the same graph.



Both curves are plotted on a log scale, the log₁₀ of the actual rate being used on the y-axis. The x-axis has no real significance; it just shows the four points corresponding to the four basic structures: MeBr, MeCH₂Br, Me₂CHBr, and Me₃CBr.



The values are also summarized in the table below, which gives the relative rates compared with that of the secondary halide, *i*-PrBr, set at 1.0 for each reaction.

Rates of S_N1 and S_N2 reactions of simple alkyl bromides

alkyl bromide type	CH ₃ Br methyl	CH ₃ CH ₂ Br primary	(CH ₃) ₂ CHBr secondary	(CH ₃) ₃ CBr tertiary
k_1 (s ⁻¹)	0.6	1.0	26	10 ⁸
$10^5 k_2$ (M ⁻¹ dm ⁻³ s ⁻¹)	13,000	170	6	0.0003
relative k_1	2×10^{-2}	4×10^{-2}	1	4×10^6
relative k_2	6×10^3	30	1	5×10^{-5}

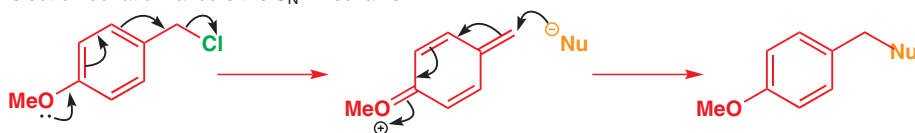
Although the reactions were chosen to give as much S_N1 reaction as possible in one case and as much S_N2 reaction as possible in the other, of course you will understand that we cannot prevent the molecules doing the 'wrong' reaction! The values for the ' S_N1 ' reaction of MeBr and MeCH₂Br are actually the low rates of S_N2 displacement of the bromide ion by the weak nucleophile HCO₂H, while the ' S_N2 ' rate for *t*-BuBr may be the very small rate of ionization of *t*-BuBr in acetone.

A closer look at electronic effects

We mentioned above that adjacent π systems increase the rate of the S_N2 reaction by stabilizing the transition state, and likewise increase the rate of S_N1 reactions by stabilizing the carbocation. The effect on the S_N2 reaction applies to both C=C (electron-rich) and C=O (electron-deficient) π systems, but only C=C π systems increase the rate of S_N1 reactions. Adjacent C=O groups in fact significantly decrease the reactivity of alkyl halides towards S_N1 reactions because the electron-withdrawing effect of the carbonyl group greatly destabilizes the carbocation.

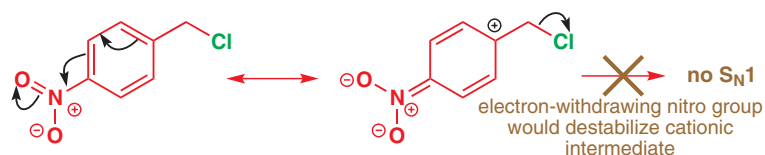
Electron-withdrawing or -donating groups can also tip finely balanced cases from one mechanism to another. For example, benzylic compounds react well by either S_N1 or S_N2 , and a change of solvent, as just discussed, might switch them from one mechanism to another. Alternatively, a benzylic compound that has a well-placed electron-donating group able to stabilize the cation will also favour the S_N1 mechanism. Thus 4-methoxybenzyl chloride reacts by S_N1 for this reason: here we show the methoxy group stabilizing the cation intermediate by assisting departure of the chloride.

electron donation favours the S_N1 mechanism

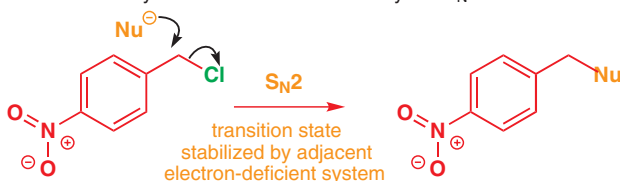


On the other hand, an electron-withdrawing group, such as a nitro group, within the benzylic compound will decrease the rate of the S_N1 reaction and allow the S_N2 mechanism to take over.

electron withdrawal disfavours the S_N1 mechanism



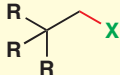
the same benzylic chloride instead reacts by the S_N2 mechanism

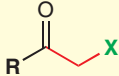
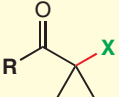


Rate measurements for benzylic chlorides illustrate the importance of this effect. We can force them all to react by S_N1 by using methanol as the solvent (methanol is a poor nucleophile and a polar solvent: both disfavour S_N2). Comparing with the rate of substitution of benzyl chloride itself, PhCH₂Cl, 4-methoxybenzyl chloride reacts with methanol about 2500 times faster and the 4-nitrobenzyl chloride about 3000 times more slowly.

● Summary of structural variations and nucleophilic substitution

We are now in a position to summarize the structural effects on both mechanisms we have been discussing over the last few pages. The table lists the structural types and rates each reaction qualitatively.

Electrophile					
	methyl	primary	secondary	tertiary	'neopentyl'
S_N1 mechanism?	bad	bad	poor	excellent	bad
S_N2 mechanism?	excellent	good	poor	bad	bad

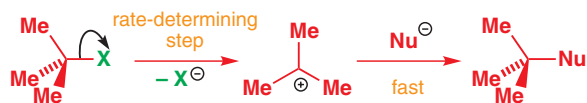
Electrophile					
	allylic	benzylic	α -alkoxy (adj. lone pair)	α -carbonyl	α -carbonyl and tertiary
S_N1 mechanism?	good	good	good	bad	bad
S_N2 mechanism?	good	good	okay but S_N1 better	excellent	possible

We have considered the important effects of the basic carbon skeleton and of solvent on the course of S_N1 and S_N2 reactions and we shall now look at two final structural factors: the nucleophile and the leaving group. We shall tackle the leaving group first because it plays an important role in both S_N1 and S_N2 reactions.

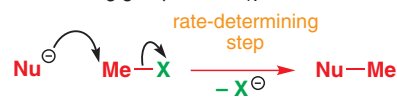
The leaving group in S_N1 and S_N2 reactions

The leaving group is important in both S_N1 and S_N2 reactions because departure of the leaving group is involved in the rate-determining step of both mechanisms.

The leaving group in the S_N1 reaction



The leaving group in the S_N2 reaction



So far you have mostly seen halides and water (from protonated alcohols) as leaving groups. Leaving groups involving halides or oxygen atoms are by far the most important, and now we need to establish the principles that make for good and bad leaving groups. As a chemist, we want leaving groups to have some staying power, so that our compounds are not too unstable, but we also don't want them to outstay their welcome—they must have just the right level of reactivity.

Halides as leaving groups

With halide leaving groups two main factors are at work: the strength of the C–halide bond and the stability of the halide ion. The strengths of the C–X bonds can be measured easily, but how can we measure anion stability? One way, which you met in Chapter 8, was to use the pK_a values of the acids HX. pK_a quantifies the stability of an anion relative to its conjugate acid. We want to know about the stability of an anion relative to that anion bonded to C, not H, but pK_a will do as a guide.

Halide leaving groups in the S_N1 and S_N2 reactions

Halide (X)	Strength of C–X bond, kJ mol ⁻¹	p <i>K</i> _a of HX
fluorine	118	+3
chlorine	81	-7
bromine	67	-9
iodine	54	-10

■ You were given the same message in Chapter 10 in relation to substitutions at C=O: hydroxide is never a leaving group. There is one exception to this rule, in the E1cb reaction, which you will meet in Chapter 17, but it's rare enough to ignore at this stage.

The table in the margin shows both bond strengths and p*K*_a. It is clearly easiest to break a C–I bond and most difficult to break a C–F bond. Iodide sounds like the best leaving group. We get the same message from the p*K*_a values: HI is the strongest acid, so it must ionize easily to H⁺ and I⁻. This result is quite correct—iodide is an excellent leaving group and fluoride a very bad one, with the other halogens in between.

Nucleophilic substitutions on alcohols: how to get an OH group to leave

Now what about leaving groups joined to the carbon atom by a C–O bond? There are many of these but the most important are OH itself, the carboxylic esters, and the sulfonate esters. First we must make one thing clear: alcohols themselves do *not* react with nucleophiles. In other words, OH⁻ is never a leaving group. Why not? For a start hydroxide ion is very basic, and if the nucleophile were strong enough to displace hydroxide ion it would be more than strong enough to remove the proton from the alcohol.

S_N2 displacement of hydroxide never happens...

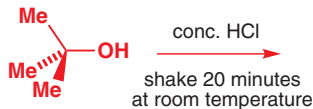


If the nucleophile reacts, it attacks the *proton* instead

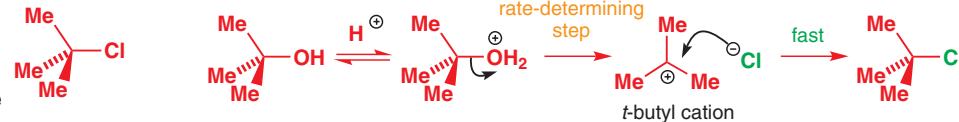


But we do want to use alcohols in nucleophilic substitution reactions because they are easily made (by the reactions in Chapter 9, for example). The simplest solution is to protonate the OH group with strong acid. This will work only if the nucleophile is compatible with strong acid, but many are. The preparation of *t*-BuCl from *t*-BuOH simply by shaking it with concentrated HCl is a good example. This is obviously an S_N1 reaction with the *t*-butyl cation as intermediate.

t-butyl chloride from *t*-butanol

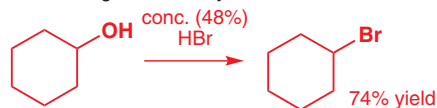


the mechanism

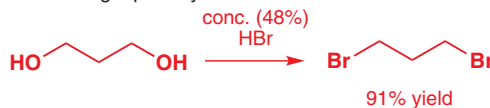


Similar methods can be used to make secondary alkyl bromides with HBr alone and primary alkyl bromides using a mixture of HBr and H₂SO₄.

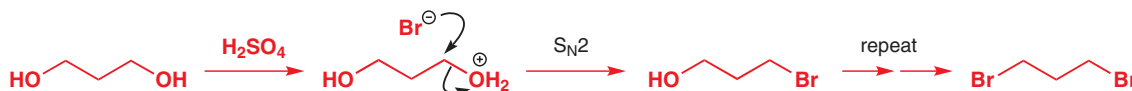
substituting a secondary alcohol in acid



substituting a primary alcohol in acid

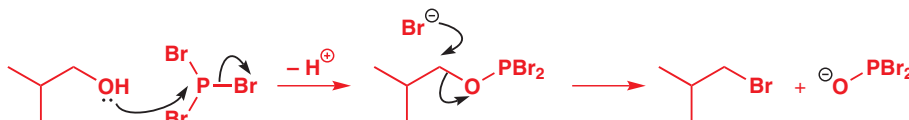
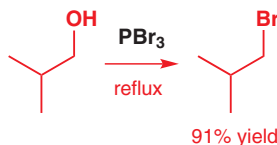


The second of these two reactions must be S_N2 , with substitution of the protonated hydroxyl group by bromide.



Another way to approach the substitution of OH is to make it a better leaving group by combination with an element that forms very strong bonds to oxygen. The most popular choices are phosphorus and sulfur. Making primary alkyl bromides with PBr₃ usually works well.

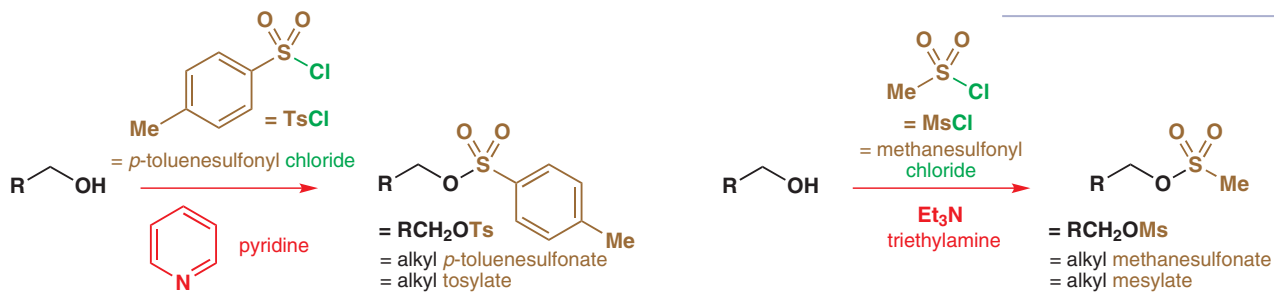
The phosphorus reagent is first attacked by the OH group (an S_N2 reaction at phosphorus) and the displacement of an oxyanion bonded to phosphorus is now a good reaction because of the anion stabilization by phosphorus.



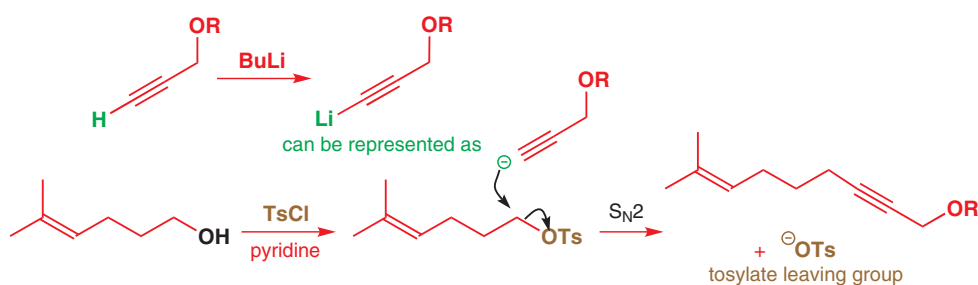
Sulfonate esters—tosylates and mesylates—from alcohols

The most widely used way of making a hydroxyl group into a good leaving group is to make it into a sulfonate ester. Primary and secondary alcohols are easily converted to sulfonate esters by treating with sulfonyl chlorides and base. The sulfonate esters are often crystalline, and are so widely used that they have been given trivial names—*tosylates* for *p*-toluenesulfonates and *mesylates* for methanesulfonates—and the functional groups have been allocated the ‘organic element’ symbols Ts and Ms.

Tosylates (*p*-toluenesulfonates) are made by treating alcohols with *p*-toluenesulfonyl chloride (or tosyl chloride) in the presence of pyridine. A similar reaction (but with a different mechanism, which we will discuss in Chapter 17) with methanesulfonyl chloride (mesyl chloride) gives a mesylate (methanesulfonate).



Sulfonic acids RSO_3H are strong acids (pK_a around 0) and so any sulfonate RSO_3^- is a good leaving group: tosylates and mesylates can be displaced by almost anything. As you saw in Chapter 8, the lithium derivative of an alkyne can be prepared by deprotonation with the very strong base butyllithium. In the example below, the tosyl derivative of a primary alcohol reacts with this lithium derivative in an S_N2 reaction. Note that the tosylate leaving group is represented as TsO^- (not Ts^- !).

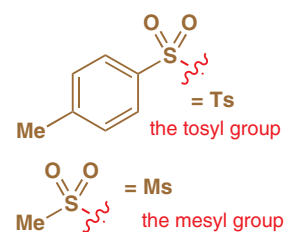
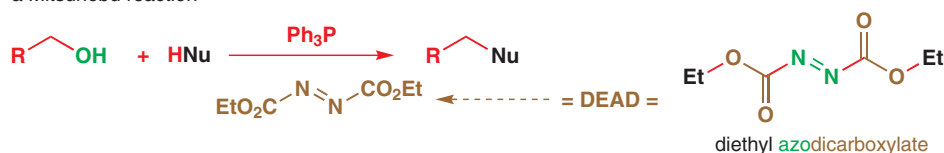


On p. 344 you saw a tosylate (we just called it a sulfonate ester then) being displaced by acetate in an S_N2 reaction. Acetate is not a very good nucleophile, and it is a testament to the power of the sulfonate esters that they are willing to act as leaving groups even with acetate, which is usually too weak to react by S_N2 .

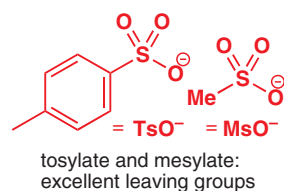
Substituting alcohols with the Mitsunobu reaction

Rather than use two steps to convert the OH group first to a sulfonate ester, and then displace it, it is possible to use a method that allows us to put an alcohol straight into a reaction mixture and get an S_N2 product in one operation. This is the Mitsunobu reaction. In this reaction, the alcohol becomes the electrophile, the nucleophile is usually relatively weak (the conjugate base of a carboxylic acid, for example), and there are two other reagents.

a Mitsunobu reaction



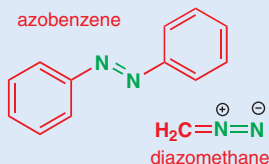
➡ The mechanism by which sulfonate esters are formed is discussed in more detail in Chapter 17.



Oyo Mitsunobu (1934–2003) worked at the Aoyama Gakuin University in Tokyo. Western chemists often misspell his name: make sure you don't!

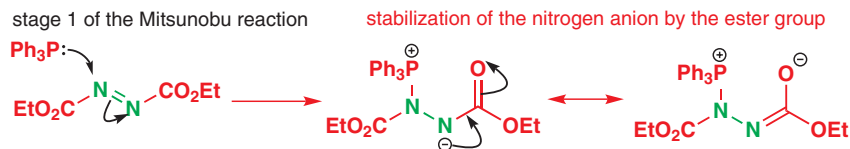
Azo compounds

The 'azo' in the name of DEAD refers to two nitrogen atoms joined together by a double bond and compounds such as azobenzene are well known. Many dyestuffs have an azo group in them—you saw some in Chapter 1. Diazo compounds, such as diazomethane, which we discuss in Chapter 38, also have two nitrogens, but only one is bonded to carbon.

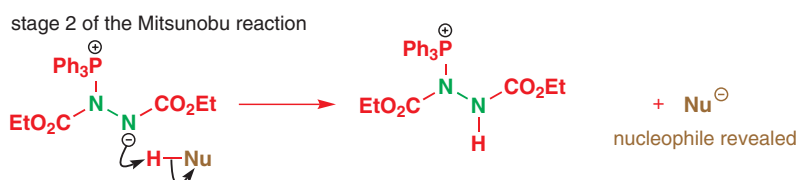


One of these reagents, Ph_3P , triphenylphosphine, is the simple phosphine you met in Chapter 11. Phosphines are nucleophilic, but not basic like amines. The other reagent deserves more comment. Its full name is diethyl azodicarboxylate, or DEAD.

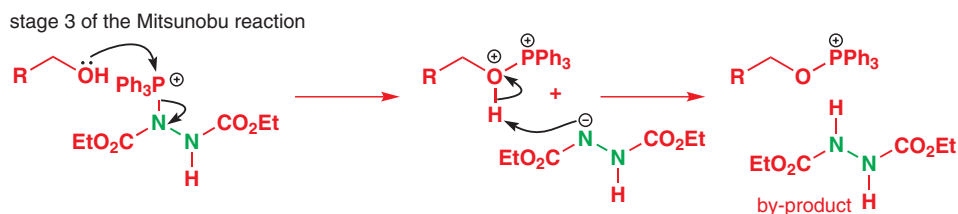
So how does the Mitsunobu reaction work? It's a long mechanism, but don't be discouraged: there is a logic to each step and we will guide you through it gently. The first stage involves neither the alcohol nor the added nucleophile. The phosphine adds to the weak $\text{N}=\text{N}$ π bond to give an anion stabilized by one of the ester groups.



You will note that the nucleophile has been added as its conjugate acid 'HNu'—often this might be a carboxylic acid, for example benzoic acid. The anion produced by this first stage is basic enough to remove a proton from this acid, generating Nu^- ready for reaction.



Oxygen and phosphorus have a strong affinity, as we saw in the conversion of alcohols to bromides with PBr_3 (p. 348) and in the Wittig reaction (Chapter 11, pp. 237–8), and the positively charged phosphorus is now attacked by the alcohol, displacing a second nitrogen anion in an $\text{S}_{\text{N}}2$ reaction at phosphorus. The nitrogen anion generated in this step is stabilized by conjugation with the ester, but rapidly removes the proton from the alcohol to give an electrophilic $\text{R}-\text{O}-\text{PPh}_3^+$ species and a by-product, the reduced form of DEAD.



Finally, the anion of the nucleophile can now attack this phosphorus derivative of the alcohol in a normal $\text{S}_{\text{N}}2$ reaction at carbon with the phosphine oxide as the leaving group. We have arrived at the products.



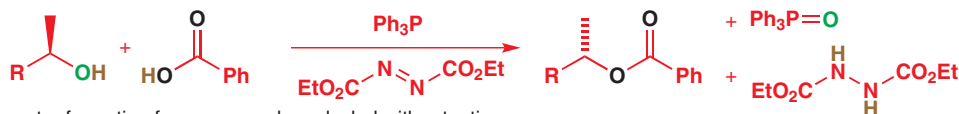
Interactive mechanism for the Mitsunobu reaction

The whole process takes place in one operation. The four reagents are all added to one flask and the products are the phosphine oxide, the reduced azo diester with two NH bonds replacing the $\text{N}=\text{N}$ double bond, and the product of an $\text{S}_{\text{N}}2$ reaction on the alcohol. Another way to look at this reaction is that a molecule of water must formally be lost: OH must be removed from the alcohol and H from the nucleophile. These atoms end up in very stable molecules—the $\text{P}=\text{O}$ and $\text{N}-\text{H}$ bonds are strong where the $\text{N}=\text{N}$ bond was weak, compensating for the sacrifice of the strong $\text{C}-\text{O}$ bond in the starting alcohol.

If this is all correct, then the vital $\text{S}_{\text{N}}2$ step should lead to inversion as it always does in $\text{S}_{\text{N}}2$ reactions. This turns out to be one of the great strengths of the Mitsunobu reaction—it is a

reliable way to replace OH by a nucleophile with inversion of configuration. The most dramatic example is probably the formation of esters from secondary alcohols with inversion. Normal ester formation leads to retention as the C–O bond of the alcohol is not broken: compare these two reactions and note the destination of the coloured oxygen (and hydrogen) atoms.

ester formation from a secondary alcohol with inversion by the Mitsunobu reaction

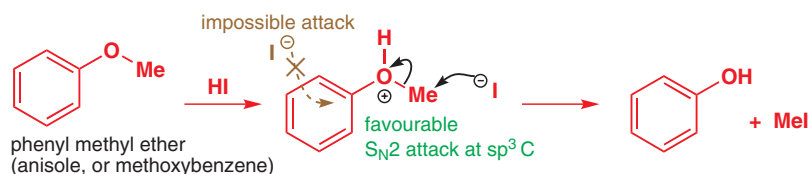


ester formation from a secondary alcohol with retention



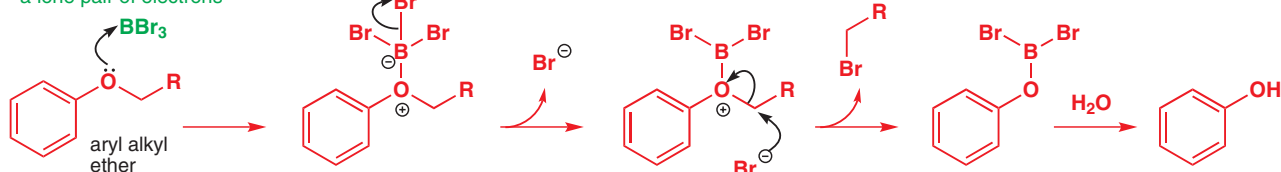
Ethers as electrophiles

Ethers are stable molecules that do not react with nucleophiles: THF and Et₂O are widely used as solvents for this reason. To make them react, we need to make the oxygen positively charged so that it can accept electrons more readily, and we also need to use a very good nucleophile. A good way of doing both is to treat with HBr or HI, which protonate the oxygen. Iodide and bromide are excellent nucleophiles in S_N2 reactions (see below), and attack will occur preferentially at the carbon atom more susceptible to S_N2 reactions (usually the less hindered one). Aryl alkyl ethers cleave only on the alkyl side—you cannot get attack through the benzene ring.



So far we have used only protic acids to help oxygen atoms to leave. But Lewis acids—species other than H⁺ that also have an empty orbital capable of accepting a lone pair—work well too, and the cleavage of aryl alkyl ethers with BBr₃ is a good example. Trivalent boron compounds have an empty p orbital so they are very electrophilic and prefer to attack oxygen. The resulting oxonium ion can be attacked by Br[−] in an S_N2 reaction.

BBr₃ acts as a Lewis acid—empty p orbital accepts a lone pair of electrons



➡ Lewis acids were introduced in Chapter 8, p. 180.

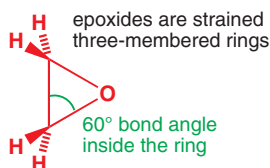
Epoxides as electrophiles

One family of ethers reacts in nucleophilic substitution even without protic or Lewis acids. They are the three-membered cyclic ethers called epoxides (or oxiranes). The leaving group is genuinely an alkoxide anion RO[−], so obviously some special feature must be present in these ethers making them unstable. This feature is ring strain, which comes from the angle between the bonds in the three-membered ring that has to be 60° instead of the ideal tetrahedral angle of 109°. You could subtract these numbers and say that there is '49° of strain' at each carbon atom, making about 150° of strain in the molecule. This is a lot: the molecule would be much

➡ You will see how to make epoxides from alkenes in Chapter 19.

more stable if the strain were released by opening up to restore the ideal tetrahedral angle at all atoms. This can be done by one nucleophilic attack.

➡ Ring strain is discussed further in Chapter 16 on p. 368.

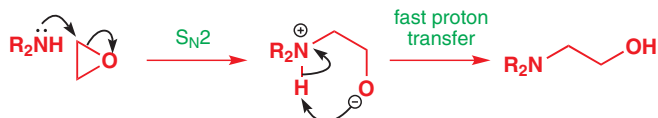


S_N2 attack on epoxides relieves ring strain

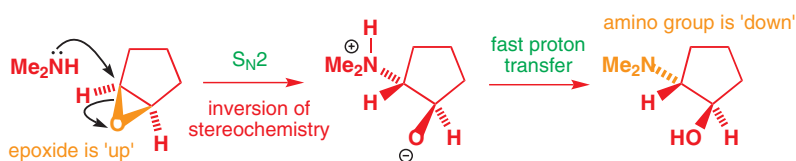


Epoxides react cleanly with amines to give amino alcohols. We have not so far featured amines as nucleophiles because their reactions with alkyl halides are often bedeviled by overreaction (see the next section), but with epoxides they give good results.

■ When epoxides are substituted differently at either end, the nucleophile has a choice of which end to attack. The factors that control this will be discussed in Chapter 24.



It is easy to see that inversion occurs in these S_N2 reactions when the epoxide is attached to (or 'fused with') another ring. With this five-membered ring nucleophilic attack with inversion gives the *trans* product. As the epoxide in the starting material is *up*, attack has to come from underneath. The new C–N bond is *down* and inversion has occurred.

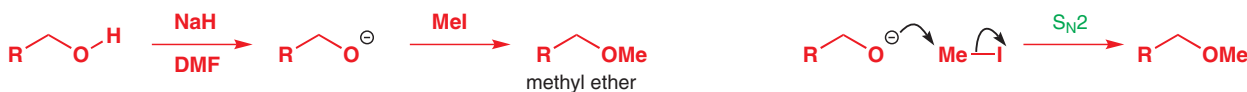


The nucleophile in S_N1 reactions

We established earlier that in an S_N1 reaction the nucleophile is not important with regard to *rate*. The rate-determining step of the reaction is loss of the leaving group, so good and bad nucleophiles all give products. We don't need to deprotonate the nucleophile to make it more reactive (water and hydroxide work just as well as each other) and this means that S_N1 reactions are often carried out under acidic conditions, to assist departure of a leaving group.

Compare, for example, these typical conditions used to make a methyl ether and a *tert*-butyl ether. The methyl ether is made, as you saw on p. 340, using methyl iodide in an S_N2 reaction. It needs a good nucleophile, so the alcohol is deprotonated to make an alkoxide with sodium hydride in DMF, which, as you saw on p. 345, is a good solvent for S_N2 reactions. The *tert*-butyl ether on the other hand is made simply by stirring the alcohol with *tert*-butanol and a little acid. No base is needed, and the reaction proceeds rapidly to give the *tert*-butyl ether.

making a methyl ether



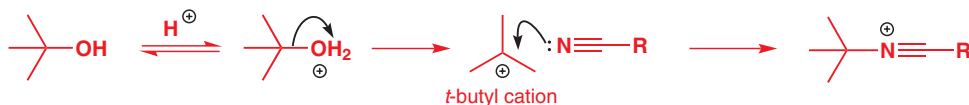
making a *tert*-butyl ether



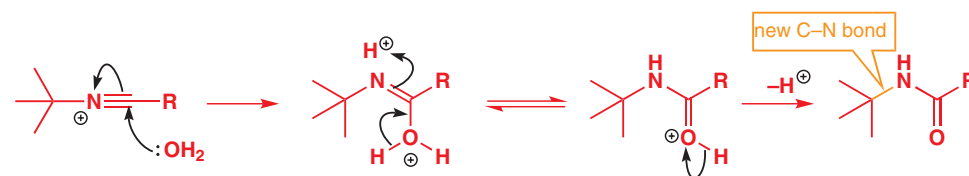
A very bad nucleophile in a good S_N1 reaction: the Ritter reaction

An interesting result of the unimportance of the nucleophile to the rate (and therefore the usefulness) of an S_N1 reaction is that very poor nucleophiles indeed may react in the absence of anything better. Nitriles, for example, are very poorly basic and nucleophilic because the lone pair of electrons on the nitrogen atom is in a low-energy sp orbital. However, if *t*-butanol is dissolved in a nitrile as solvent and strong acid is added, a reaction does take place. The acid does not protonate the nitrile, but does protonate the alcohol to produce the *t*-butyl cation in the usual first step of an S_N1 reaction. This cation is reactive enough to combine with even such a weak nucleophile as the nitrile.

nitriles are very weak bases and poor nucleophiles



The resulting cation is captured by the water molecule released in the first step and an exchange of protons leads to a secondary amide. The overall process is called the Ritter reaction and it is one of the few reliable ways to make a C–N bond to a tertiary centre.

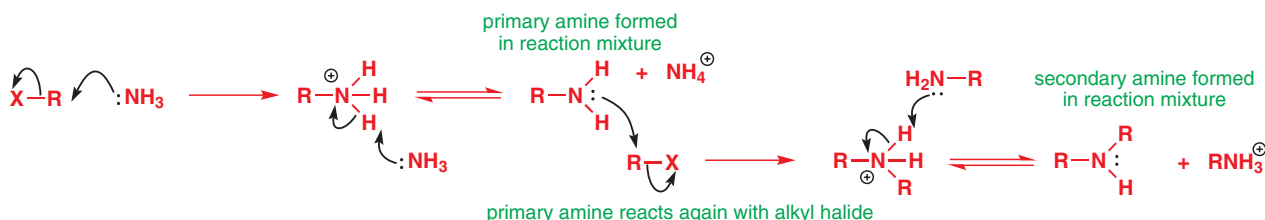


The nucleophile in the S_N2 reaction

In an S_N2 reaction, a good nucleophile is essential. We finish this chapter with a survey of effective choices for forming new bonds to sp³ by S_N2 reactions, and a description of the factors that determine how good a nucleophile will be.

Nitrogen nucleophiles: a problem and a solution

Amines are good nucleophiles, but reactions between ammonia and alkyl halides rarely lead cleanly to single products. The problem is that the product of the substitution is at least as nucleophilic as the starting material, so it competes for reaction with the alkyl halide.

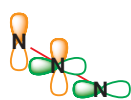


Even this is not all! The alkylation steps keep going, forming the secondary and tertiary amines, and stopping only when the non-nucleophilic tetra-alkylammonium ion R₄N⁺ is formed. The problem is that the extra alkyl groups push more and more electron density onto N, making each product more reactive than the previous. The quaternary ammonium salt could probably be made cleanly if a large excess of alkyl halide RX is used, but other more controlled methods are needed for the synthesis of primary, secondary, and tertiary amines.

One solution for primary amines is to replace ammonia with azide ion N₃[−]. This linear triatomic species, nucleophilic at both ends, is a slender rod of electrons able to insert itself into almost any electrophilic site. It is available as the water-soluble sodium salt NaN₃.

Sometimes these alkylations can work, but usually only if the alkylating agent or the amine is very hindered, or the alkylating agent contains an inductive electron-withdrawing group (such as the hydroxyl group generated when an epoxide is opened: epoxides are reliable alkylating agents for amines). With amine alkylations, you should nonetheless always expect the worst.

structure of azide ion N_3^-



■ Azide is isoelectronic with carbon dioxide, and has the same linear shape.

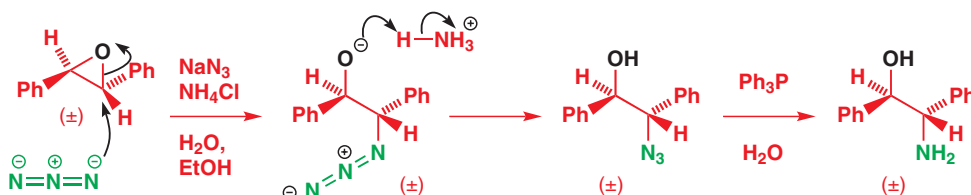
Azide reacts only once with alkyl halides because the product, an alkyl azide, is no longer nucleophilic. However, rarely is the azide product required: it is usually reduced to a primary amine by catalytic hydrogenation (H_2 over a Pd catalyst—see Chapter 23), LiAlH_4 , or triphenylphosphine.

A warning about azides

Azides can be converted by heat—or even sometimes just by a sharp blow—suddenly into nitrogen gas. In other words they are potentially explosive, particularly inorganic (that is, ionic) azides and low molecular weight covalent organic azides.



Azides react with epoxides too. This epoxide is one diastereoisomer (*trans*) but racemic and the symbol ($\pm$) under each structure reminds you of this (Chapter 14). Azide attacks at either end of the three-membered ring (the two ends are the same) to give the hydroxy-azide. The reaction is carried out in a mixture of water and an organic solvent with ammonium chloride as buffer to provide a proton for the intermediate. Triphenylphosphine in water is used for reduction to the primary amine.



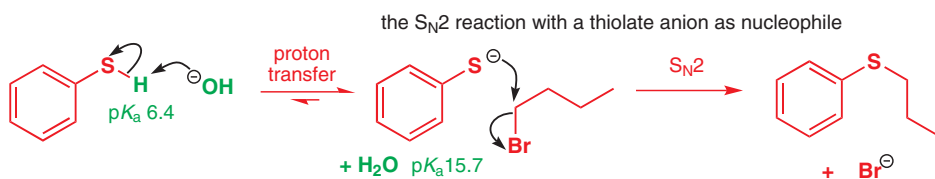
➔ The mechanism of the reduction of azides by triphenylphosphine can be found on p. 1176.

Sulfur nucleophiles are better than oxygen nucleophiles in $\text{S}_{\text{N}}2$ reactions

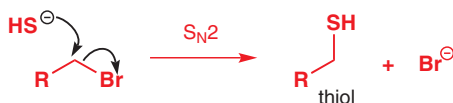
Thiolate anions RS^- make excellent nucleophiles in $\text{S}_{\text{N}}2$ reactions on alkyl halides. It is enough to combine the thiol, sodium hydroxide, and the alkyl halide to get a good yield of the sulfide.



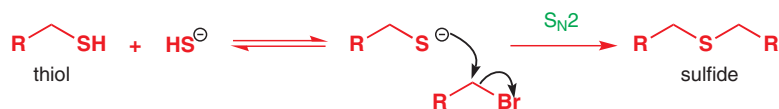
Thiols are more acidic than water (pK_{a} of RSH is typically 9–10, pK_{a} of PhSH is 6.4, pK_{a} of H_2O is 15.7) and rapid proton transfer from sulfur to oxygen gives the thiolate anion that acts as a nucleophile in the $\text{S}_{\text{N}}2$ reaction.



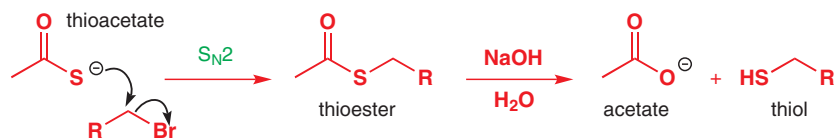
But how do you make a thiol in the first place? The obvious way to make aliphatic thiols would be by an $\text{S}_{\text{N}}2$ reaction using NaSH on the alkyl halide.



This works well but, unfortunately, the product easily exchanges a proton and the reaction normally produces the symmetrical sulfide—this should remind you of what happened with amines!



The solution is to use the anion of thioacetic acid, usually the potassium salt. This reacts cleanly through the more nucleophilic sulfur atom and the resulting ester can be hydrolysed in base to liberate the thiol.



Effectiveness of different nucleophiles in the S_N2 reaction

In Chapter 10 we pointed out that basicity is nucleophilicity towards protons. At that stage we said that nucleophilicity towards the carbonyl group parallels basicity almost exactly. We are able to use pK_a as a guide to the effectiveness of nucleophilic substitution reactions at the **carbonyl group**.

During this chapter you have had various hints that nucleophilicity towards saturated carbon is not so straightforward. Now we must look at this question seriously and try to give you helpful guidelines.

1. If the atom that is forming the new bond to carbon is the same over a range of nucleophiles—it might be oxygen, for example, and the nucleophiles might be HO[−], PhO[−], AcO[−], and TsO[−]—then nucleophilicity does parallel basicity. The anions of the weakest acids are the best nucleophiles. The order for the nucleophiles we have just mentioned will be: HO[−] > PhO[−] > AcO[−] > TsO[−]. The actual values for the rates of attack of the various nucleophiles on MeBr in EtOH relative to the rate of reaction with water (= 1) are given in the table below.

Relative rates (water = 1) of reaction with MeBr in EtOH

Nucleophile X [−]	pK _a of HX	Relative rate
HO [−]	15.7	1.2 × 10 ⁴
PhO [−]	10.0	2.0 × 10 ³
AcO [−]	4.8	9 × 10 ²
H ₂ O	−1.7	1.0
ClO ₄ [−]	−10	0

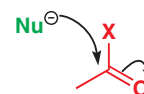
2. If the atoms that are forming the new bond to carbon are *not* the same over the range of nucleophiles we are considering, then another factor is important. In the very last examples we have been discussing we have emphasized that RS[−] is an excellent nucleophile for saturated carbon. Let us put that another way: RS[−] is a better nucleophile for saturated carbon than RO[−], even though RO[−] is more basic than RS[−] (see table below).

Relative rates (water = 1) of reaction with MeBr in EtOH

Nucleophile X [−]	pK _a of HX	Relative rate
PhS [−]	6.4	5.0 × 10 ⁷
PhO [−]	10.0	2.0 × 10 ³

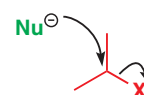
Sulfur is plainly a better nucleophile than oxygen for saturated carbon. Why should this be? As we discussed back in Chapter 5, there are two main factors controlling bimolecular reactions: (1) electrostatic attraction (simple attraction of opposite charges or partial charges) and (2) bonding interactions between the HOMO of the nucleophile and the LUMO of the electrophile.

pK_a of HNu is a good guide to the rate of this sort of reaction



nucleophilic attack on C=O

but the story with this sort of reaction is more complicated

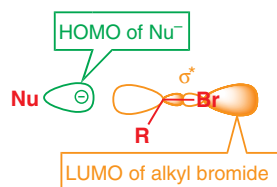


nucleophilic substitution at saturated C

A proton is, of course, positively charged, so electrostatic attraction is the more important factor in nucleophilicity towards H^+ , or pK_a . The carbonyl group too has a substantial positive charge on the carbon atom, arising from the uneven distribution of electrons in the $\text{C}=\text{O}$ π bond, and reactions of nucleophiles with carbonyl groups are also heavily influenced by electrostatic attraction, with HOMO–LUMO interactions playing a smaller role.

Electronegativities:

C: 2.55 I: 2.66 Br: 2.96 O: 3.44

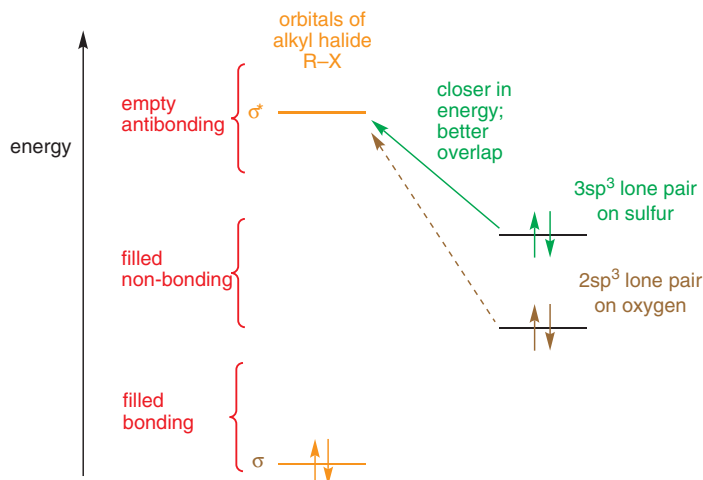


When it comes to saturated carbon atoms carrying leaving groups, polarization is typically much less important. There is, of course, some polarity in the bond between a saturated carbon atom and, say, a bromine atom, but the electronegativity difference between C and Br is less than half that between C and O. In alkyl iodides, one of the best classes of electrophiles in $\text{S}_\text{N}2$ reactions, there is in fact almost no dipole at all—the electronegativity of C is 2.55 and that of I is 2.66.

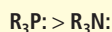
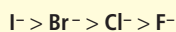
● Electrostatic attraction is often unimportant in $\text{S}_\text{N}2$ reactions.

What does matter is the strength of the HOMO–LUMO interaction. In a nucleophilic attack on the carbonyl group, the nucleophile adds in to the low-energy π^* orbital. In a nucleophilic attack on a saturated carbon atom, the nucleophile must donate its electrons to the σ^* orbital of the $\text{C}-\text{X}$ bond, as illustrated in the margin for an alkyl bromide reacting with the non-bonding lone pair of a nucleophile.

σ^* antibonding orbitals are, of course, higher in energy than non-bonding lone pairs, but the higher the energy of the nucleophile's lone pair, the better the overlap. The 3sp^3 lone-pair electrons of sulfur overlap better with the high-energy σ^* orbital of the $\text{C}-\text{X}$ bond than do the lower energy 2sp^3 lone-pair electrons on oxygen because the higher energy of the sulfur electrons brings them closer in energy to the $\text{C}-\text{X}$ σ^* orbital. The conclusion is that nucleophiles from lower down the periodic table are more effective in $\text{S}_\text{N}2$ reactions than those from the top rows.



● Typically, nucleophilic power towards saturated carbon goes like this:



Nucleophiles in substitution reactions

Relative rates (water = 1) of reaction of nucleophiles with MeBr in EtOH

nucleophile	F^-	H_2O	Cl^-	Et_3N	Br^-	PhO^-	EtO^-	I^-	PhS^-
relative rate	0.0	1.0	1100	1400	5000	2.0×10^3	6×10^4	1.2×10^5	5.0×10^7

Hard and soft nucleophiles

The fact that some nucleophiles, like $\text{R}_3\text{P:}$ and RS^- , react very fast at saturated C atoms (they have high-energy lone pairs), but very poorly at C=O groups (they are either uncharged or have charge spread diffusely over large orbitals) gives them a different type of character from strongly basic nucleophiles like HO^- that attack C=O groups rapidly. We call nucleophiles that react well at saturated carbon **soft** nucleophiles; those that are more basic and react well with carbonyl groups are referred to as **hard** nucleophiles. These are useful and evocative terms because the soft nucleophiles are indeed rather large and flabby with diffuse high-energy electrons while the hard nucleophiles are small and spiky with closely held electrons and high charge density.

When we say 'hard' (nucleophile or electrophile) we refer to species whose reactions are dominated by electrostatic attraction and when we say 'soft' (nucleophile or electrophile) we refer to species whose reactions are dominated by HOMO–LUMO interactions.

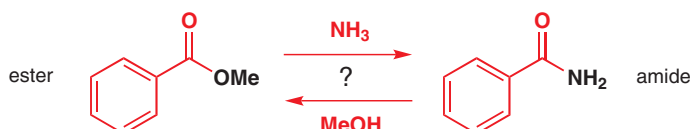
■ Just to remind you: reactions dominated by electrostatic attraction also need to pass electrons from HOMO to LUMO, but reactions that are dominated by HOMO–LUMO interactions need have no contribution from electrostatic attraction.

● Summary of the characteristics of the two types of nucleophile.

Hard nucleophiles X	Soft nucleophiles Y
small	large
charged	neutral
basic (HX weak acid)	not basic (HY strong acid)
low-energy HOMO	high-energy HOMO
like to attack C=O	like to attack saturated carbon
such as RO^- , NH_2^- , MeLi	such as RS^- , I^- , R_3P

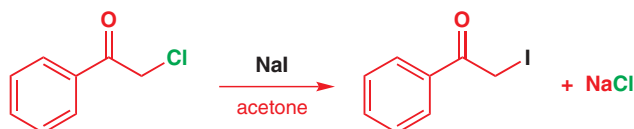
Nucleophiles and leaving groups compared

In Chapter 10 we explained that, in a nucleophilic attack on the carbonyl group, a good nucleophile is a bad leaving group and vice versa. We set you the challenge of predicting which way the following reaction would go.

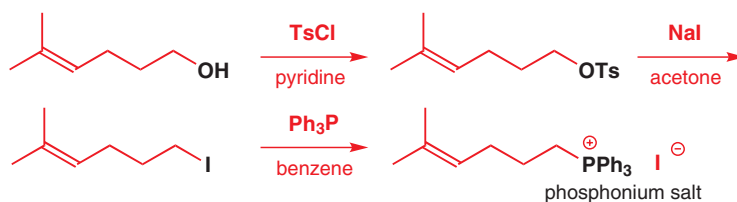


You should by now understand well that the reaction goes from ester to amide rather than the other way round, because NH_3 is a better nucleophile than MeOH and NH_2^- is a worse leaving group than MeO^- .

The $\text{S}_{\text{N}}2$ reaction is different: some of the best nucleophiles are also the best leaving groups. The most important examples of this are bromide and iodide. As the table on p. 356 showed, iodide ion is one of the best nucleophiles towards saturated carbon because it is at the bottom of its group in the periodic table and its lone-pair electrons are very high in energy. Alkyl iodides are readily formed from alkyl chlorides or tosylates. Here are two examples. The first is assisted by the solvent, acetone, which allows NaCl to precipitate and drives the reaction forward.

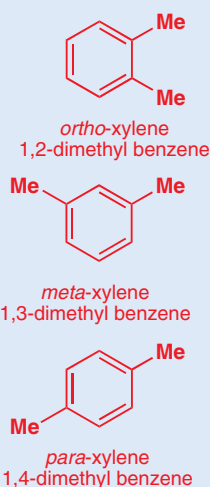


The second example is from the preparation of a phosphonium salt used in a synthesis of terpenes. An unsaturated primary alcohol was first made into its tosylate, the tosylate was converted into the iodide, and the iodide into the phosphonium salt.



■ We explained on p. 347 why I^- is such a good leaving group: the C–I bond is particularly weak. The poor overlap between the atomic orbitals on C and on I also mean that the σ^* is low lying and easily accessible to the nucleophile's HOMO.

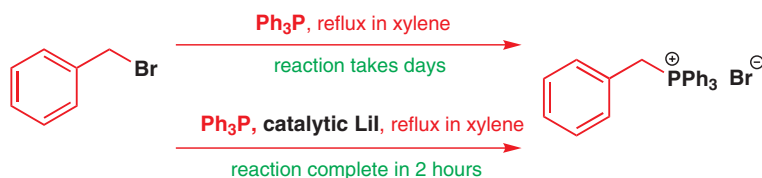
The solvent 'xylene' needs some explanation. Xylene is the trivial name for dimethyl benzene and there are three isomers. Mixed xylenes are isolated cheaply from oil and often used as a relatively high boiling solvent (b.p. about 140°C) for reactions at high temperature. In this case, the starting materials are soluble in xylene but the product is a salt and conveniently precipitates out during the reaction. Non-polar xylene favours the $\text{S}_{\text{N}}2$ reaction (p. 345).



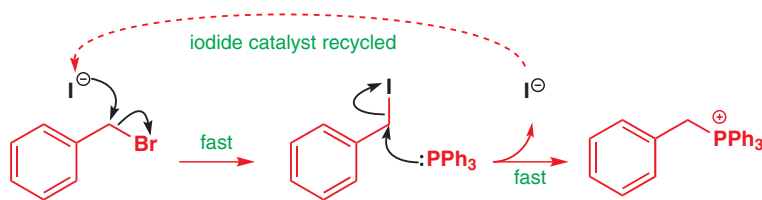
A nucleophilic catalyst speeds up a reaction by acting as both a good nucleophile and a good leaving group. You saw pyridine performing a similar function in substitution reactions at the $\text{C}=\text{O}$ group of acid anhydrides in Chapter 10.

But why this roundabout route via the iodide? The answer is that as well as being an excellent nucleophile, iodide is such a good leaving group that alkyl iodides are often used as intermediates to encourage substitution with other nucleophiles. Yields are often higher if the alkyl iodide is prepared than if the eventual nucleophile is reacted directly with the alkyl tosylate or chloride.

However, iodine is expensive, and a way round that problem is to use a catalytic amount of iodide. The phosphonium salt below is formed slowly from benzyl bromide but the addition of a small amount of LiI speeds up the reaction considerably.



The iodide reacts both as a better nucleophile than Ph_3P and then as a better leaving group than Br^- . Each iodide ion goes round the cycle many times as a **nucleophilic catalyst**.



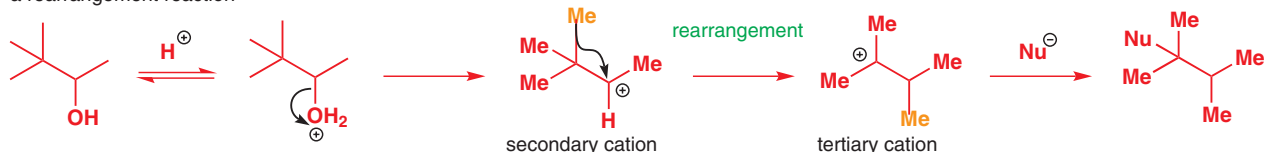
Looking forward: elimination and rearrangement reactions

Simple nucleophilic substitutions at saturated carbon atoms are fundamental reactions found wherever organic chemistry is practised. They are used in industry on an enormous scale and in pharmaceutical laboratories to make important drugs. They are worth studying for their importance and relevance.

There is another side to this simple picture. These were among the first reactions whose mechanisms were thoroughly investigated by Ingold in the 1930s and since then they have probably been studied more than any other reactions. All our understanding of organic mechanisms begins with $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions, and you need to understand these basic mechanisms properly.

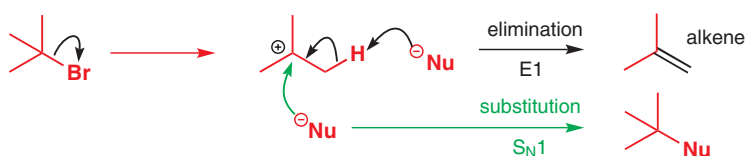
The carbocations you met in this chapter are reactive intermediates not only in $\text{S}_{\text{N}}1$ substitutions but in other reactions too. One of the most convincing pieces of evidence for their formation is that they undergo reactions other than simple addition to nucleophiles. For example, the carbon skeleton of the cation may *rearrange*, as we will discuss in Chapter 36.

a rearrangement reaction



Another common fate of cations, and something that may also happen instead of an intended S_N1 or S_N2 reaction, is *elimination*. Here an alkene is formed by the nucleophile acting as a base to remove HX instead of adding to the molecule.

an elimination reaction (E1)



You will meet elimination reactions in the next chapter but one (17) after some further exploration of stereochemistry.

Further reading

This subject is treated in every organic chemistry textbook, often as the first reaction described. Good examples include: J. Keeler and P. Wothers, *Why Chemical Reactions Happen*, OUP, Oxford, 2003,

chapter 11 and F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry A, Structure and Mechanisms*, 5th edn, Springer, 2007, chapter 4.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>